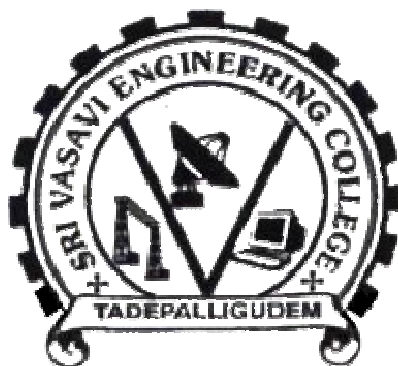


ACADEMIC REGULATIONS COURSE STRUCTURE and DETAILED SYLLABUS COMPUTER SCIENCE

For

M.Tech., TWO YEAR P.G COURSE
(Applicable for batches admitted from 2021-2022)



•08818-284577, 284355 Ext: 321; Fax: 08818-284577

Visit us at: www.srivasaviengg.ac.in

SRI VASAVI ENGINEERING COLLEGE
(Autonomous)

(Sponsored by Sri Vasavi Educational Society)

(Approved by AICTE, New Delhi & Permanently affiliated to JNTUK, Kakinada)

(Accredited by NAAC with A Grade ,Recognized by UGC under section 2(f) & 12(B))

Pedatadepalli, **TADEPALLIGUDEM – 534 101. W.G.Dist. (A.P)**

Institute Vision and Mission

Vision:

To be a premier technological institute striving for excellence with global perspective and commitment to the nation.

Mission:

- To produce engineering graduates of professional quality and global perspective through Learner Centric Education.
- To establish linkages with government, industry and research laboratories to promote R&D activities and to disseminate innovations.
- To create an eco-system in the institute that leads
- to holistic development and ability for life-long learning.

Department Vision and Mission

Vision:

To evolve as a centre of academic and research excellence in the area of Computer Science and Engineering.

Mission :

- To utilize innovative learning methods for academic improvement.
- To encourage higher studies and research to meet the futuristic requirements of Computer Science and Engineering.
- To inculcate Ethics and Human values for developing students with good character.

**ACADEMIC RULES
AND
REGULATIONS**

ACADEMIC REGULATIONS V21 FOR M. Tech DEGREE COURSE

(Applicable for the batch of students admitted from the Academic Year 2021-2022)

Applicable for the students of M. Tech (Regular) programme from the Academic Year 2020-21 onwards. The M. Tech Degree from Sri Vasavi Engineering college, Tadepalligudem shall be conferred on candidates by the affiliating university who are admitted to the program and who fulfill all the requirements for the award of the Degree.

1.0 ELIGIBILITY FOR ADMISSIONS

Admission to the above program shall be made subject to eligibility, qualification and specialization as prescribed by the University/ State Government from time to time.

Admissions shall be made on the basis of merit/rank obtained by the candidates at the qualifying Entrance Test conducted by the University or on the basis of any other order of merit as approved by the University/ State Government, subject to reservations as laid down by the Govt. from time to time.

2.0 AWARD OF M. Tech DEGREE

2.1 A student shall be declared eligible for the award of the M. Tech Degree, if he pursues program of study in not less than two and not more than four academic years.

2.2 The student shall register for all 68 credits and secure all the 68 credits.

2.3 The minimum instruction days in each semester are 90.

3.0 PROGRAMME OF STUDY

The following specializations are offered at present for the M. Tech Programme of study.

CE	M.Tech. – CIVIL(Structural Engineering)
EEE	M.Tech- Power Electronics and Power Systems
ME	M.Tech- Thermal Engineering
ECE	M.Tech- Embedded System & VLSI
CSE	M.Tech- Computer Science

and any other course as approved by AICTE/ University from time to time.

4.0 ATTENDANCE

- 4.1 A student shall be eligible to appear for examinations if he acquires a minimum of 75% of attendance in aggregate of all the courses, and with minimum 50% in each and every course including practicals.
- 4.2 Condonation of shortage of attendance in aggregate up to 10% (65% and above and below 75%) in each semester shall be granted by the College Academic Committee.
- 4.3 Shortage of Attendance **below** 65% in aggregate shall not be condoned.
- 4.4 Students whose shortage of attendance is not condoned in any semester are not eligible to write their end semester examination of that semester.
- 4.5 A prescribed fee shall be payable towards condonation of shortage of attendance.
- 4.6 A student shall not be promoted to the next semester unless, he satisfies the attendance requirement of the present semester, as applicable. They may seek re-admission into that semester when offered next. If any candidate fulfills the attendance requirement in the present semester, he shall not be eligible for re-admission into the same class.

5.0 EVALUATION

The performance of the candidate in each semester shall be evaluated course-wise, with a maximum of 100 marks for theory and practical, on the basis of Internal Evaluation and End Semester Examination.

- 5.1 For the theory courses 70 marks shall be awarded based on the performance in the End Semester Examination and 30 marks shall be awarded based on the continuous Internal Evaluation. The internal evaluation shall be made based on the **average** of the marks secured in the two Mid Term-Examinations conducted-one in the middle of the Semester and the other immediately after the completion of instruction. Each mid term examination shall be conducted for a total duration of 120 minutes with 3 questions (without choice) each for 10 marks. End semester examination is conducted for 70 marks for all FIVE (5) questions (one question from one unit) to be answered (either or).
- 5.2 For practical courses, 70 marks shall be awarded based on the performance in the End Semester Examinations and 30 marks shall be awarded based on the day-to-day performance as Internal Marks. The internal evaluation based on the day to day work-10 marks, record- 10 marks and the remaining 10 marks to be awarded by conducting an internal laboratory test. The end examination shall be conducted by the examiners, with a breakup marks of Procedure-20, Experimentation-20, Results-20, Viva-voce-10.
- 5.3 For Mini Project with Seminar, a student under the supervision of a faculty member, shall collect the literature on a topic and critically review the literature and submit it to the department in a report form and shall make an oral presentation before the Project Review Committee consisting

of Head of the Department, supervisor/mentor and two other senior faculty members of the department. For Mini Project with Seminar, there will be only internal evaluation for 50 marks. A candidate has to secure a minimum of 50% of marks to be declared successful.

- 5.4** A candidate shall be deemed to have secured the minimum academic requirement in a course if he secures a minimum of 40% of marks in the End semester Examination and a minimum aggregate of 50% of the total marks in the End Semester Examination and Internal Evaluation taken together.
- 5.5** In case the candidate does not secure the minimum academic requirement in any course (as specified in 5.4) he has to re-appear for the End semester Examination in that course. A candidate shall be given **one** chance to re-register for each course provided the internal marks secured by a candidate **are less than 50% and has failed in the end examination**. In such a case, the candidate must re-register for the course(s) and secure the required minimum attendance. The candidate's attendance in the re-registered course(s) shall be calculated separately to decide upon his eligibility for writing the end examination in those course(s). In the event of the student taking another chance, his internal marks and end examination marks obtained in the previous attempt shall stand cancelled. For re-registration the candidates have to apply to the college by paying the requisite fees and get approval before the start of the semester in which re-registration is required.
- 5.6** In case the candidate secures less than the required attendance in any re-registered course(s), he shall not be permitted to write the End Examination in that course. He shall again re-register the course when next offered.
- 5.7** Laboratory examination for M. Tech. courses must be conducted with two Examiners, one of them being the Laboratory Class Teacher and the second examiner shall be appointed by the principal from the panel of examiners submitted by the respective department.

6.0 EVALUATION OF PROJECT/DISSERTATION WORK

Every candidate shall be required to submit a thesis or dissertation on a topic approved by the Project Review Committee.

- 6.1** The work on the project shall be initiated at the beginning of the II year and the duration of the project is two semesters. A candidate is permitted to submit Project Thesis only after successful completion of theory and practical course with the approval of PRC not earlier than 40 weeks from the date of registration of the project work. The candidate has to pass all the theory and practical courses before submission of the Thesis.
- 6.2** A Project Review Committee (PRC) shall be constituted with Head of the Department and two other senior faculty members in the department.
- 6.3** Registration of Dissertation/Project Work: A candidate is permitted to register for the project work after satisfying the attendance requirement of all the courses, both theory and practical upto 2nd semester
- 6.4** After satisfying 6.3, a candidate has to submit, in consultation with his project supervisor, the title, objective and plan of action of his project work for approval. The student can initiate the Project work, only after obtaining

the approval from the Project Review Committee (PRC).

- 6.5 If a candidate wishes to change his supervisor or topic of the project, he can do so with the approval of the Project Review Committee (PRC). However, the PRC shall examine whether or not the change of topic/supervisor leads to a major change of his initial plans of project proposal. If yes, his date of registration for the project work will be taken as the date on which the change of Supervisor or topic as the case may be applied.
- 6.6 Continuous assessment of Dissertation-I and Dissertation-II during the Semester(s) will be monitored by the PRC.
- 6.7 A candidate shall submit his status report in two stages to the PRC, at least with a gap of 3 months between them.
- 6.8 At the end of the III semester project phase-1 is evaluated for 50 marks by the committee casting of HOD, supervisor and external examiner appointed by principal.
- 6.9 Three copies of the Project Thesis certified by the supervisor shall be submitted to the department.
- 6.10 The thesis shall be adjudicated by one examiner selected by the principal. For this, the HOD of the department shall submit a panel of 5 examiners, eminent in that field, with the help of the guide concerned.
- 6.11 The Head of the Department shall coordinate and make arrangements for the conduct of Viva-Voce examination.
- 6.12 If the report of the examiner is favorable, Viva-Voce examination shall be conducted by a board consisting of the Supervisor, Head of the Department and the examiner who adjudicated the Thesis. The Board shall jointly evaluate the candidate's work for a maximum of 100 marks.
- 6.13 If the report of the examiner is not favourable, the candidate shall revise and resubmit the Thesis, in the time frame as decided by the PRC. If the report of the examiner is unfavorable again, the thesis shall be summarily rejected. The candidate has to re-register for the project and complete the project within the stipulated time after taking the approval from the principal.
- 6.14 If the report of the Viva-Voce is unsatisfactory (ie, < 50 marks), the candidate shall retake the Viva-Voce examination only after three months. If he fails to get a satisfactory report at the second Viva-Voce examination, the candidate has to re-register for the project and complete the project within the stipulated time after taking the approval from the University.

7.0 Supplementary Examinations

Supplementary examinations will be conducted twice in a year at the end of odd and even semester.

8.0 Revaluation

Recounting of marks in the end semester examinations

- 8.1 As per the notification issued by the controller of examinations, the students can submit the application for revaluation, along with the requisite fee receipt for revaluation of his/her answer script(s) of the theory course(s), if he /she is not satisfied with the marks obtained.
- 8.2 The Controller of examinations shall arrange for re-evaluate the answer script(s).

- 8.3 A new Examiner, other than the first examiner, shall re-evaluate the answer script(s).
- 8.4 Better marks out of the two shall be taken into consideration.
- 8.5 If the difference of marks between the two valuations is more than 15%, the answer script will be referred to third valuation. The average of nearest two marks will be awarded.

9.0 Cumulative Grade Point Average (CGPA)

Marks Range Theory/ Laboratory (Max - 100)	Letter Grade	Level	Grade Point
≥ 90%	O	Outstanding	10
≥80 to <90%	S	Excellent	9
≥70 to <80%	A	Very Good	8
≥60 to <70%	B	Good	7
≥50 to <60%	C	Satisfactory	6
<50%	F	Fail	0
		Absent	0

Computation of SGPA

- The following procedure will be adopted to compute the Semester Grade Point Average (SGPA) and Cumulative Grade Point Average (CGPA):

The **SGPA** is the ratio of sum of the product of the number of credits with the grade points scored by a student in all the courses taken by a student and the sum of the number of credits of all the courses undergone by a student, i.e. $SGPA (S_i) = \sum (C_i \times G_i) / \sum C_i$

- Where C_i is the number of credits of the i th course and G_i is the grade point scored by the student in the i th course.

Computation of CGPA

- The **CGPA** is also calculated in the same manner taking into account all the courses undergone by a student over all the semester of a Programme, i.e.
- $CGPA = \sum (C_i \times S_i) / \sum C_i$**
- Where S_i is the SGPA of the i th semester and C_i is the total number of credits in that semester.
- The SGPA and CGPA shall be rounded off to 2 decimal points and reported in the transcripts.
- Equivalent Percentage = $(CGPA - 0.75) \times 10$

10.0 AWARD OF DEGREE AND CLASS

After a student has satisfied the requirements prescribed for the completion of the program and is eligible for the award of M. Tech. Degree he shall be placed in one of the following four classes:

	CGPA to be secured	
First Class with Distinction	≥ 7.75	From the CGPA secured from 68 Credits.
First Class	≥ 6.75	
Second Class	≥ 5.75 to < 6.75	
Pass Class	≥ 4.75 to < 5.75	

The Grades secured, Grade points and Credits obtained will be shown separately in the memorandum of marks.

11.0 WITHHOLDING OF RESULTS

If the student is involved in indiscipline/malpractices/court cases, the result of the student will be withheld.

12.0 TRANSITORY REGULATIONS (for V21)

12.1 When a student gets detained due to academic regulations and rejoins the college to complete the programme. However, the academic regulations under which he/she was first admitted shall continue to be applicable to him/her.

12.2 When a student discontinues for some time and rejoins the college to complete the programme. However, the academic regulations under which he/she was first admitted shall continue to be applicable to him/her.

13.0 GENERAL

13.1 Wherever the words –he||, –him||, –his||, occur in the regulations, they include –she||, –her||, –hers||.

13.2 The academic regulation should be read as a whole for the purpose of any interpretation.

13.3 In the case of any doubt or ambiguity in the interpretation of the above rules, the decision of the Vice-Chancellor is final.

13.4 The University may change or amend the academic regulations or syllabi at any time and the changes or amendments made shall be applicable to all the students with effect from the dates notified by the University.

COURSE STRUCTURE

M.Tech(CS) Programme Course Structure

(With effect from 2021-22 Admitted Batch onwards)

SEMESTER-I

S.No.	Course Code	Course	L	T	P	C
1	V21CTT01	Program Core-1 Mathematical Foundations of Computer Science	3	-	-	3
2	V21CTT02	Program Core-2 Advanced Data Structures	3	-	-	3
3	Program Elective-I		3	-	-	3
	V21CTT03	1. Advanced Operating Systems				
	V21CTT04	2. Advanced Computer Architecture				
	V21CTT05	3. Parallel Computing				
4	Program Elective-II		3	-	-	3
	V21CTT06	1. Advanced Databases				
	V21CTT07	2. Advanced Computer Networks				
	V21CTT08	3. Object Oriented Software Engineering				
5	V21MBT55	Research Methodology and IPR	2	-	-	2
6	V21CTL01	Laboratory-1 Advanced Data Structures Lab	-	-	4	2
7	Laboratory-2: Advanced Computing Lab-1(<i>Lab programs based on elective taken by student may be offered</i>)		-	-	4	2
	V21CTL02	Advanced Operating Systems				
	V21CTL03	Parallel Computing				
	V21CTL04	Advanced Computer Networks				
	V21CTL05	Object Oriented Software Engineering				
8		Audit Course-1*	2	-	-	0
Total Credits			16	0	8	18

Student has to choose any one audit course listed below*SEMESTER-II**

S.No.	Course Code	Course	L	T	P	C
1	V21CTT09	Program Core-3 Web Technologies	3	-	-	3
2	V21CTT10	Program Core-4 Data Science through Python Programming	3	-	-	3
3	Program Elective-III		3	-	-	3
	V21CTT11	1. Machine Learning				
	V21CTT12	2. Ad hoc and Sensor Networks				
	V21CTT13	3. Internet of Things				
4	Program Elective-IV		3	-	-	3
	V21CTT14	1. Principles of Cyber Security				
	V21CTT15	2. Cloud Computing				
	V21CTT16	3. Natural Language Processing				
7	V21CTL06	Advanced Web Technologies Lab	-	-	4	2
8	V21CTL07	Data Science Applications with Python Lab	-	-	4	2
9	V21CTM01	Mini Project with Seminar	2	-	-	2
10		Audit Course-2*	2	0	0	0
Total Credits			16	0	8	18

**Student has to choose any one audit course listed below*

**Student has to choose any one audit course listed below.*

Audit Course 1 & 2:

1. English for Research Paper Writing
2. Disaster Management
3. Value Education
4. Stress Management by Yoga
5. Personality Development through Life Enlightenment Skills
6. Pedagogy Studies

M.Tech(CS) Programme Course Structure

(With effect from 2021-22 Admitted Batch onwards)

SEMESTER-III

S.No.	Course Code	Course	L	T	P	C
1	Program Elective-V		3	-	-	3
	V21CTT17	1. MOOCS-1 through NPTEL/ SWAYAM ¹ 12 Week Program related to the programme which is not listed in the course structure				
	V21CTT18	2. Mobile Applications and Development				
	V21CTT19	3. Big Data Analytics				
2	Open Elective		3	-	-	3
	V21CTT20	MOOCs-2 Through NPTEL /SWAYAM - Any 12 week course on Engineering/ Management/ Mathematics offered by other than parent department				
	V21MAT02	Operations Research				
	V21MBT56	Cost Management of Engineering Projects				
3	V21CTP01	Dissertation-I/Industrial Project #	-	-	-	10
Total Credits						16

#Students going for Industrial Project/Thesis will complete these courses through MOOCs

SEMESTER-IV

S.No.	Course Code	Course	L	T	P	C
1	V21CTP02	Dissertation-II	-	-	-	16
Total Credits						16

SYLLABUS

I SEM	Mathematical Foundations of Computer Science	Course Code: V21CTT01	L	T	P	C
			3	0	0	3

Course Outcomes: After completion of course, students would be able to

- CO1:** Demonstrate skills in solving mathematical problems, mathematical principles and logic. (K3)
CO2: Demonstrate the basic concepts associated with set theory, relations, functions and their applications (K3)
CO3: Illustrate algebraic structures and concepts associated with Number Theory and their applications in computer science. (K3)
CO4: Manipulate and consider data numerically by using combinatorics. (K3)
CO5: Solve recurrence relations using various methods apply techniques of graphs for real-time problems(K3)

UNIT I: **Mathematical Logic**-Propositional Calculus: Statements and Notations, Connectives, Well Formed Formulas, Truth Tables, Tautologies, Equivalence of Formulas, Duality Law, Tautological Implications, Normal Forms, Theory of Inference for Statement Calculus, Consistency of Premises, Indirect Method of Proof. Predicate Calculus: Predicative Logic, Statement Functions, Variables and Quantifiers, Free and Bound Variables, Inference Theory for Predicate Calculus.

UNIT II: **Set Theory**-Introduction, Operations on Binary Sets, Principle of Inclusion and Exclusion, *Relations*: Properties of Binary Relations, Relation Matrix and Digraph, Operations on Relations, Partition and Covering, Transitive Closure, Equivalence, Compatibility and Partial Ordering Relations, Hasse Diagrams, *Functions*: Bijective Functions, Composition of Functions, Inverse Functions, Permutation Functions, Recursive Functions, Lattice and its Properties.

UNIT III: **Algebraic Structures and Number Theory**- *Algebraic Structures*: Algebraic Systems, Examples, General Properties, Semi Groups and Monoids, Homomorphism of Semi Groups and Monoids, Group, Subgroup, Abelian Group, Homomorphism, Isomorphism, *Number Theory*: Properties of Integers, Division Theorem, The Greatest Common Divisor, Euclidean Algorithm, Least Common Multiple, Testing for Prime Numbers, The Fundamental Theorem of Arithmetic, Modular Arithmetic (Fermat's Theorem and Euler's Theorem).

UNIT IV: **Combinatorics**- Basic of Counting, Permutations, Permutations with Repetitions, Circular Permutations, Restricted Permutations, Combinations, Restricted Combinations, Generating Functions of Permutations and Combinations, Binomial and Multinomial Coefficients, Binomial and Multinomial Theorems, The Principles of Inclusion-Exclusion, Pigeonhole Principle and its Application.

UNIT V: **Recurrence Relations**-Generating Functions, Function of Sequences, Partial Fractions, Calculating Coefficient of Generating Functions, Recurrence Relations, Formulation as Recurrence Relations, Solving Recurrence Relations by Substitution and Generating Functions, Method of Characteristic Roots, Solving Inhomogeneous Recurrence Relations, **Graph Theory**: Basic Concepts of Graphs, Sub graphs, Matrix Representation of Graphs: Adjacency Matrices, Incidence Matrices, Isomorphic Graphs, Paths and Circuits, Eulerian and Hamiltonian Graphs, Multigraphs, Planar Graphs, Euler's Formula, Graph Colouring and Covering, Chromatic Number, Spanning Trees, Algorithms for Spanning Trees (Problems Only and Theorems without Proofs).

Text Books:

1. Discrete Mathematical Structures with Applications to Computer Science, J.P. Tremblay and P. Manohar, Tata McGraw Hill.
2. Elements of Discrete Mathematics-A Computer Oriented Approach, C. L. Liu and D. P. Mohapatra, 3rd Edition, Tata McGraw Hill.
3. Discrete Mathematics and its Applications with Combinatorics and Graph Theory, K. H. Rosen, 7th Edition, Tata McGraw Hill.

Reference Books:

1. Discrete Mathematics for Computer Scientists and Mathematicians, J. L. Mott, A. Kandel, T. P. Baker, 2nd Edition, Prentice Hall of India.
2. Discrete Mathematical Structures, Bernand Kolman, Robert C. Busby, Sharon Cutler Ross, PHI.
3. Discrete Mathematics, S. K. Chakraborty and B.K. Sarkar, Oxford, 2011.

I SEM	Advanced Data Structures	Course Code: V21CTT02	L	T	P	C
			3	0	0	3

Course Outcomes: After completion of course, students would be able to

- CO1:** Select appropriate data structures as applied to specified problem definition. (K2)
CO2: Apply data structures such as linked list and trees on necessary applications. (K3)
CO3: Practice all data structures like stacks, queues, trees, graphs and compare their performance. (K3)
CO4: Discuss operations like searching, insertion, deletion and traversing on various data structures. (K2)
CO5: Apply data structures into the applications such as binary search trees, AVL, Red black trees. (K3)

UNIT I: **Arrays** Abstract Data Types and the C++ Class, The Array as an Abstract Data Type, The Polynomial Abstract Data type, Sparse Matrices, Introduction- Sparse Matrix Representation- Transposing a Matrix- Matrix Multiplication, Representation of Arrays. **Stacks And Queues-** Templates in C++, The Stack Abstract Data Type, The Queue Abstract Data Type, Subtyping and Inheritance in C++, Evaluation of Expressions, Expression- Postfix Notation- Infix to Postfix.

UNIT II: **Linked Lists** Single Linked List and Chains, Representing Chains in C++, The Template Class Chain, Circular Lists, Available Space Lists, Linked Stacks and Queues, Polynomials, Equivalence Classes, Sparse Matrices, Doubly Linked Lists, Generalized Lists, Representation of Generalized Lists, **Trees** Introduction, Binary Trees, Binary Tree Traversal and Tree Iterators- Introduction, In order, Preorder, Post order Traversal, Thread Binary Trees, Heaps, Binary Search Trees.

UNIT III: **Graphs** The Graph Abstract Data Type, Elementary Graph Operation, Minimum Cost Spanning Trees, Shortest Paths and Transitive Closure, **Hashing**-Introduction, Static Hashing, Dynamic Hashing.

UNIT IV: **Priority Queues** Binomial heaps, Fibonacci Heaps, Symmetric Min- Max Heaps, **Efficient Binary Search Trees** Optimal Binary Search Trees, AVL trees, Red-Black Trees, Splay Trees.

UNIT V: **Multiway Search Trees** m-way Search Trees, B- Trees, B+- Trees **Digital Search Trees** Digital Search Trees, Binary Tries and Patricia, Multiway Tries

Text Books:

1. Data structures, Algorithms and Applications in C++, S.Sahni, 2nd edition, Universities Press, Pvt. Ltd.
2. Data structures and Algorithm Analysis in C++, Mark Allen Weiss, Pearson Education. Ltd., Second Edition.
3. Data structures and Algorithms in C++, Michael T.Goodrich, R.Tamassia and Mount, Wiley student edition, John Wiley and Sons.

Reference Books:

1. Data structures and algorithms in C++, 3rd Edition, Adam Drozdek, Thomson
2. Data structures using C and C++, Langsam, Augenstein and Tanenbaum, PHI.
3. Problem solving with C++, The OOP, Fourth edition, W.Savitch, Pearson education.

I SEM	Advanced Operating Systems (Elective-I)	Course Code: V21CTT03	L	T	P	C
			3	0	0	3

Course Outcomes: After completion of course, students would be able to

- CO1:** Explain architectures and issues in Distributed Operating Systems.(K2)
CO2: Illustrate different Distributed Mutual Exclusion Algorithms and Distributed Deadlock Algorithms. (K3)
CO3: Demonstrate Distributed Scheduling Algorithm and Distributed Shared Memory.(K3)
CO4: Apply various Cryptographic Algorithms for the protection of given data.(K3)
CO5: Demonstrate Multiprocessor Scheduling Algorithms and Concurrency Control Algorithms.(K3)

UNIT-I: Architectures of Distributed Systems: System Architecture Types, Distributed Operating Systems, Issues in Distributed Operating Systems. **Theoretical Foundations:** Inherent Limitations of a Distributed System, Lamport's Logical clocks, Vector Clocks, Global State, and Termination Detection.

UNIT-II: Distributed Mutual Exclusion: The classification of Mutual Exclusion Algorithms, Preliminaries, Non-Token-Based Algorithms, Lamport's Algorithm, The Ricart-Agrawala Algorithm, Token-Based Algorithms, Suzuki-kasami's Broadcast Algorithm, Singhal's Heuristic Algorithm. **Distributed Deadlock Detection:** Resource Vs Communication Deadlocks, A graph- theoretic Model, Deadlock handling strategies in Distributed Systems, Control Organizations for Distributed Deadlock Detection, Centralized Deadlock Detection Algorithms, Distributed Deadlock Detection Algorithms- A Path-Pushing, Edge-Chasing, Hierarchical Deadlock Detection Algorithms. **Agreement protocols:** The System Model, The Byzantine Agreement Problem, The Consensus Problem.

UNIT-III: Distributed File Systems: Mechanisms for Building Distributed File Systems, Design Issues. **Distributed Shared Memory:** Algorithms for Implementing DSM, Memory Coherence, Coherence Protocols, Design Issues. **Distributed Scheduling:** Issues in Load Distributing, Components of a Load Distributing Algorithm, Stability, Load Distributing Algorithms, Performance Comparison.

UNIT-IV: Failure Recovery: Backward and Forward Error Recovery, Consistent Set of Checkpoints, Synchronous and Asynchronous check Pointing and Recovery. **Fault Tolerance:** Commit Protocols, Non-blocking Commit Protocols, Voting Protocols. **Resource Protection and Security:** The Access Matrix Model. **Cryptography:** Private Key: Data Encryption Standard, Public Key Cryptography.

UNIT-V: Multiprocessor System Architectures: Basic Multiprocessor System Architectures, Interconnection Networks for Multiprocessor Systems, Caching, Hypercube Architectures. **Multiprocessor Operating Systems:** Threads, Process Synchronization, Processor Scheduling. **Database Operating Systems:** Concurrency Control Model of Database systems, Problem of Concurrency Control, Distributed Database Systems. **Concurrency Control Algorithms:** Lock Based Algorithms, Timestamp Based Algorithms.

TEXT BOOKS:

1. "Advanced concepts in operating systems: Distributed, Database and multiprocessor operating systems", Mukesh Singhal, Niranjana G.Shivaratri, TMH, 2001.

REFERENCE BOOKS:

1. "Modern operating system", Andrew S.Tanenbaum, PHI, 2003
2. "Distributed operating system-Concepts and design", Pradeep K.Sinha, PHI, 2003.
3. "Distributed operating system", Andrew S.Tanenbaum, Pearson education, 2003.

I SEM	Advanced Computer Architecture (Elective-I)	Course Code: V21CTT04	L	T	P	C
			3	0	0	3

Course Outcomes: After completion of course, students would be able to

- CO1:** Classify the types of computers, and new trends and developments in computer architecture. (K2)
CO2: Describe pipelining, instruction set architectures, memory addressing. (K2)
CO3: Demonstrate exploiting ILP using dynamic scheduling, multiple issue, and speculation. (K3)
CO4: Demonstrate the various techniques to enhance a processors ability to exploit Instruction level parallelism (ILP), and its challenges. (K3)
CO5: Illustrate multithreading by using ILP and supporting thread-level parallelism (TLP). (K3)

UNIT I: Fundamentals of Computer Design-Fundamentals of Computer design, Changing faces of computing and task of computer designer, Technology trends, Cost price and their trends, Measuring and reporting performance, Quantitative principles of computer design, Amdahl's law, Instruction set principles and examples- Introduction, Classifying instruction set- Memory addressing- type and size of operands, Operations in the instruction set.

UNIT II: Pipelines- Introduction, Basic RISC instruction set, Simple implementation of RISC instruction set, Classic five stage pipe lined RISC processor, Basic performance issues in pipelining, Pipeline hazards, Reducing pipeline branch penalties, **Memory Hierarchy Design-** Introduction, Review of ABC of cache, Cache performance, Reducing cache miss penalty, Virtual memory.

UNIT III: Instruction Level Parallelism the Hardware Approach: Instruction- Level parallelism, Dynamic scheduling, Dynamic scheduling using Tomasulo's approach, Branch prediction, high performance instruction delivery- hardware based speculation.

UNIT IV: ILP Software
Approach Basic compiler level techniques, Static branch prediction, VLIW approach, Exploiting ILP, Parallelism at compile time, Cross cutting issues - Hardware versus Software.

UNIT V: Multi Processors and Thread Level Parallelism:
Multi Processors and Thread level Parallelism- Introduction, Characteristics of application domain, Systematic shared memory architecture, Distributed shared-memory architecture, Synchronization, **Inter Connection and Networks-** Introduction, Interconnection network media, Practical issues in interconnecting networks, Examples of inter connection, Cluster, Designing of clusters, **Intel Architecture-** Intel IA-64 ILP in embedded and mobile markets Fallacies and pit falls.

Text Books:

1. JohnL. Hennessy, David A.Patterson –Computer Architecture: A Quantitative Approach, 3rd Edition, An Imprint of Elsevier.

References:

1. John P.Shen and Miikko H.Lipasti – Modern Processor Design: Fundamentals of Super Scalar Processors
2. Computer Architecture and Parallel Processing – Kai Hwang, Faye A.Brigs., MC Graw Hill.
3. Advanced Computer Architecture – A Design Space Approach – Dezso Sima, Terence Fountain, Peter Kacsuk , Pearson Ed

I SEM	Parallel Computing (Elective-I)	Course Code: V21CTT05	L	T	P	C
			3	0	0	3

Course Outcomes: After completion of course, students would be able to

- CO1:** Describe different parallel architectures; inter-connect networks, programming models, and algorithms for common operations such as matrix-vector multiplication. **(K2)**
- CO2:** Develop an efficient parallel algorithm to solve it. **(K3)**
- CO3:** Illustrate a parallel algorithm time complexity as a function of the problem size and number of processors. **(K3)**
- CO4:** Illustrate Matrix Multiplication and Sorting Techniques. **(K3)**
- CO5:** Explain parallel algorithm using MPI, Open MP, pthreads, or a combination of MPI and Open MP. **(K2)**

UNIT I: **History-** Introduction, Modern Scientific Method, Evolution of Supercomputing, Modern Parallel Computers, Seeking Concurrency, Data Clustering, Programming Parallel Computers. **Parallel Architectures:** Introduction, Interconnection Networks, Processor Arrays, Multiprocessors, Multi computers, Flynn's Taxonomy

UNIT II: **Parallel Algorithm Design-** Introduction, The Task/Channel Model, Foster's Design Methodology, Boundary Value Problem, Finding the Maximum, The n-Body Problem, Adding Data Input, **Message-Passing Programming-** Introduction, The Message-Passing Model, The Message-Passing Interface, Circuit Satisfiability, Introducing Collective Communication, Benchmarking Parallel Performance.

UNIT III: **The Sieve of Eratosthenes-** Introduction, Sequential Algorithm, Sources of Parallelism, Data Decomposition options, Developing the Parallel Algorithm, Analysis of Parallel Sieve Algorithm, Documenting the Parallel Program, Benchmarking, Improvements, **Performance Analysis-** Introduction, Speedup and Efficiency, Amdahl's Law, Gustafson-Barsis's Law, The Karp-Flatt Metric, The Iso-efficiency Metric.

UNIT IV: **Matrix Multiplication,** Introduction, Sequential Matrix Multiplication, Row wise Block-Striped Parallel Algorithm, Cannon's Algorithm, Solving Linear Systems, Back Substitution, Gaussian Elimination, Iterative Methods, **Sorting** Introduction, Quick sort, A Parallel Quick sort Algorithm, Hyper quick sort Algorithm, Parallel Sorting by Regular Sampling.

UNIT V: **Shared-Memory Programming** – Introduction, The Shared-Memory Model, Parallel for Loops, Declaring Private Variables, Critical section, Reductions, Performance Improvements, More General Data Parallelism, Functional Parallelism, **Combining MPI and OpenMP** -Introduction, Conjugate Gradient Method, Jacobi Method.

Text Books:

1. Parallel Programming in C with MPI and OpenMP Michael J, Quinn Oregon State University.

Reference books:

1. Distributed and Cloud Computing: From Parallel Processing to the Internet of Things 1st Edition, Kai Hwang, Jack Dongarra, Geoffrey C. Fox

I SEM	Advanced Databases (Elective-II)	Course Code: V21CTT06	L	T	P	C
			3	0	0	3

Course Outcomes: After completion of course, students would be able to

- CO1:** Explain Distributed Database Process, Architecture, and Design Principles. **(K2)**
CO2: Apply Distributed Query Optimization Techniques and Algorithms. **(K3)**
CO3: Illustrate and apply Concurrency Control and Reliability Techniques. **(K3)**
CO4: Illustrate Need of Complex Data type like ORDBMS and OODBMS. **(K3)**
CO5: Identify Emerging Database Models and case study on Time Series Database. **(K2)**

UNIT I: Database Analysis and Design Techniques: Review of basic Database Concepts, Database Design Methodologies. ER Modeling: Specialization, Generalization, Aggregation, Normalization Theory. Database Implementation using UML: Introduction to UML, Structure diagrams, behavioral diagrams, object oriented analysis, class diagram, Advanced Transaction Processing and Concurrency Control: Transaction Concepts, Concurrency Control: Locking Methods, Time stamping Methods, Optimistic Methods for Concurrency Control, Concurrency Control in Distributed Systems.

UNIT II: Query Compiler: Introduction, parsing, generating logical query plan from parse tree. Query Processing: Physical-Query-plan Operators. Operations: selection, sorting, join, project, set. Query Evaluation: Introduction, Approaches to QE, Transformation of relational expressions in Query optimization, heuristic optimization, cost estimation for various operations, transformation rule.

UNIT III: Distributed Database Centralized DBMS and Distributed DBMS, functions and architecture of a DDBMS, Distributed Data Storage, Transparency issues in DDBMS, Query Processing DDBMS, Distributed transaction Management and Protocols, Distributed Concurrency Control and Deadlock Management.

UNIT IV: Object Oriented Database Limitations of RDBMS, Need of Complex Datatype, Data Definition, ODBMS Fundamentals, issues in OODBMS, Object- oriented database design. Comparison of ORDBMS and OODBMS.

UNIT V: Emerging Database Models, Technologies and Applications Multimedia database-Emergence, difference from other data types, structure, deductive databases, GIS and spatial databases, Knowledge database, Information Visualization, Wireless Networks and databases, Personal database, Digital libraries, web databases, case studies.

Time Series Database(TSD): Time Series Database, Importance of TSD, TSD Vs Relational DB, RTA and TSD, TSD as stream processing solution.

Text Books:

1. Advanced database management system by RiniChkrabarti and Shibhadra Dasgupta, Dreamtech.
2. Distributed Databases by Ozsu and Valduriez ,Pearson Education.

Reference Books:

1. Fundamentals of Database Systems by Ramez Elmasri, Shamkant Navathe, Pearson Education
2. Database System Concepts by Abraham Silberschatz, Henry F. Korth, S. Sudarshan, Tata McGraw-Hill.
3. <https://www.xenonstack.com/insights/time-series-databases-in-real-time-analytics>

I SEM	Advanced Computer Networks (Elective-II)	Course Code: V21CTT07	L	T	P	C
			3	0	0	3

Course Outcomes: After completion of course, students would be able to

- CO1:** Describe the functionalities and services provided by the network layer. (K2)
CO2: Apply IP addressing for the given network. (K3)
CO3: Select the transport protocol appropriate for a given application. (K3)
CO4: Discriminate between different types of multimedia communications. (K2)
CO5: Describe the working, types and challenges involved in Adhoc networks & SDN. (K2)

UNIT I: **Network layer**-Network Layer design issues: store-and forward packet switching, services provided transport layers, implementation connection less services, implementation connection oriented services, comparison of virtual – circuit and datagram subnets. **Routing Algorithm** –shortest path routing, flooding, distance vector routing, link state routing, Hierarchical routing, Broadcast routing, Multicasting routing, routing for mobiles Hosts, routing in Adhoc networks, **Congestion control algorithms**-Load shedding, Congestion control in Data gram Subnet.

UNIT II: **IPV4 Address** address space, notations, classful addressing, classless addressing network addressing translation(NAT) , **IPV6 Address** structure address space, **Internetworking** need for network layer internet as a data gram, internet as connection less network. **IPV4** datagram, Fragmentation, checksum, options. **IPV6** Advantages, packet format, extension Headers, Transition form IPV4 to IPV6

UNIT III: **Process to process delivery:** client/server paradigm, multiplexing and demultiplexing, connectionless versus connection oriented services, reliable versus reliable. **UDP:** well known ports for UDP, user data gram, check sum, UDP operation, and uses of UDP **TCP:** TCP services, TCP features, segement, A TCP connection, Flow control, error control, congestion control. **SCTP:** SCTP services SCTP features, packet format, An SCTP association, flow control, error control. **Congestioncontrol:** open loop congestion control, closedloop congestion control, Congestion control in TCP, frame relay, **Quality Of Service:** flow characteristics, flow classes **Techniques To Improve QOS:** scheduling, traffic shaping, resource reservation, admission control.

UNIT IV: **Multimedia-** introduction digital a audio , Audio compression, streaming audio, internet radio, voice over IP, introduction to video, video compression, video on demand, the Mbone-the multicast back bone

UNIT V: **Emerging trends Computer Networks- Mobile Ad hoc networks:** applications of Ad hoc networks, challenges and issues in MANETS,MAC layers issues, routing protocols in MANET, transport layer issues, Ad Hoc networks security. **Wireless sensors networks:** WSN functioning, operation system support in sensor devices, WSN Characteristics, sensor network operation, sensor Architecture: cluster management; **Wireless mesh networks** WMN design, Issues in WMNs;
Software-defined networking (SDN): importance of SDN, working principle of SDN, Benefits of SDN, SDN vs Traditional Networking, Different models of SDN.

Text Books:

1. Data communications and networking 4th edition Behrouz A Fourzan, TMH.
2. Computer networks 4th edition Andrew S Tanenbaum, Pearson.
3. Computer networks, Mayank Dave, CENGAGE.

Reference Books:

1. Computer Networks, A system Approach, 5th ed, Larry L Peterson and Bruce S Davie, Elsevier.
2. <https://www.vmware.com/topics/glossary/content/software-defined-networking>.

I SEM	Object Oriented Software Engineering (Elective-II)	Course Code: V21CTT08	L	T	P	C
			3	0	0	3

Course Outcomes: After completion of course, students would be able to

- CO1:** Apply the Object Oriented Software-Development Process to design software. (K3)
CO2: Illustrate the Object Oriented Software Architecture & Design patterns. (K3)
CO3: Examine the Design and Plan software solutions to problems using an object-oriented Testing strategies. (K4)
CO4: Illustrate the Object Oriented metrics for designing the projects. (K3)
CO5: Develop the CASE Tools & Integrated CASE environments for Object Oriented Software. (K3)

UNIT I: Introduction to Software Engineering: Software, Software Crisis, Software Engineering definition, Evolution of Software Engineering Methodologies, Software Engineering Challenges. Software Processes: Software Process, Process Classification, Phased development life cycle, Software Development Process Models- Process, use, applicability and Advantages/limitations.

UNIT II: Object oriented Paradigm, Object oriented Concepts, Classes, Objects, Attributes, Methods and services, Messages, Encapsulation, Inheritance, Polymorphism, Identifying the elements of object model, management of object oriented Software projects, Object Oriented Analysis, Domain Analysis, Generic Components of OOA model, OOA Process, Object Relationship model, Object Behavior Model, Object Oriented Design: Design for Object-Oriented systems, The Generic components of the OO design model, The System design process, The Object design process, Design Patterns, Object Oriented Programming.

UNIT III: Object Oriented testing: Broadening the view of Testing, Testing of OOA and OOD models, Object-Oriented testing strategies, Test case design for OO software, testing methods applicable at the class level, Interclass test case design

UNIT IV: Technical Metrics for Object Oriented Systems: The Intent of Object Oriented metrics, The distinguishing Characteristics, Metrics for the OO Design model, Class-Oriented metrics, Operation-Oriented Metrics, Metrics for Object Oriented testing, Metrics for Object Oriented projects

UNIT V: Computer-Aided Software Engineering: What is CASE?, Building blocks for CASE, A taxonomy of CASE tools, Integrated CASE environments, The Integration Architecture, The CASE Repository

Text Books:

1. Object Oriented and Classical Software Engineering, 7/e, Stephen R. Schach, TMH.
2. Object oriented and Classical Software Engineering, Timothy Lethbridge, Robert Laganieri, TMH
3. Software Engineering by Roger S Pressman, Tata McGraw Hill.

Reference Books:

1. Component based Software Engineering: ivica Crnkovic, Springer.

I SEM	Research Methodology and IPR	Course Code: V21MBT55	L	T	P	C
			2	0	0	2

Course Outcomes: After completion of course, students would be able to

- CO1:** Discuss different methodologies and techniques used in research work. (K2)
CO2: Explain basic computer skills necessary for the conduct of research. (K2)
CO3: Assess the basic function and working of analytical instruments used in research. (K3)
CO4: Practice the required numerical skills necessary to carry out research. (K3)
CO5: Demonstrate a capacity to identify, apply and assess ownership rights and marketing protection under intellectual property law as applicable to information, ideas, new products and product marketing. (K3)

- UNIT 1:** Meaning of research problem, Sources of research problem, Criteria Characteristics of a good research problem, Errors in selecting a research problem, Scope and objectives of research problem. Approaches of investigation of solutions for research problem, data collection, analysis, interpretation, Necessary instrumentations.
UNIT 2: Effective literature studies approaches, analysis Plagiarism, Research ethics, Effective technical writing, how to write report, Paper Developing a Research Proposal, Format of research proposal, a presentation and assessment by a review committee.
UNIT 3: Nature of Intellectual Property: Patents, Designs, Trade and Copyright. Process of Patenting and Development: technological research, innovation, patenting, development. International Scenario: International cooperation on Intellectual Property. Procedure for grants of patents, Patenting under PCT.
UNIT 4: Patent Rights: Scope of Patent Rights. Licensing and transfer of technology. Patent information and databases. Geographical Indications.
UNIT 5: New Developments in IPR: Administration of Patent System. New developments in IPR; IPR of Biological Systems, Computer Software etc. Traditional knowledge Case Studies, IPR and IITs.

REFERENCES:

- (1) Stuart Melville and Wayne Goddard, "Research methodology: an introduction for science& engineering students"
- (2) Wayne Goddard and Stuart Melville, "Research Methodology: An Introduction"
- (3) Ranjit Kumar, 2nd Edition, "Research Methodology: A Step by Step Guide for beginners"
- (4) Halbert, "Resisting Intellectual Property", Taylor & Francis Ltd ,2007.
- (5) Mayall, "Industrial Design", McGraw Hill, 1992.
- (6) Niebel, "Product Design", McGraw Hill, 1974.
- (7) Asimov, "Introduction to Design", Prentice Hall, 1962.
- (8) Robert P. Merges, Peter S. Menell, Mark A. Lemley, " Intellectual Property in New Technological Age", 2016.
- (9) T. Ramappa, "Intellectual Property Rights Under WTO", S. Chand, 2008

I SEM	Advanced Data Structures Lab	Course Code: V21CTL01	L	T	P	C
			0	0	4	2

Course Outcomes: After completion of course, students would be able to

- CO1:** Develop solutions for a range of problems using object oriented programming. (K3)
CO2: Practice complex problems using advanced data structures like arrays, stacks, queues, linked lists, graphs and trees. (K3)
CO3: Practice operations like searching, insertion, deletion and traversing on various data structures. (K3)
CO4: Apply data structures into the applications such as binary search trees, AVL, Red black trees. (K3)
CO5: Differentiate various hash functions. (K2)

List of Experiments

Experiment 1:

Develop a program to implement Polynomial additions using Arrays.

Experiment 2:

Develop a program to implement Polynomial additions using linked lists.

Experiment 3:

Develop a program to implement Multi Stack ADT using Arrays with the basic operations as Create(), IsEmpty(), Push(), Pop(), IsFull() with appropriate prototype to a functions.

Experiment 4:

Develop a program to implement Queue ADT using Linked list with the basic functions of Create(), IsEmpty(), Insert(), Delete() and IsFull() with suitable prototype to a functions.

Experiment 5:

Develop a program to transfer data from stack to queue.

Experiment 6:

Develop a program to implement the following ADT using Linked list with the basic functions of Create(), IsEmpty(), Insert(), Delete() and IsFull() with suitable prototype to a functions.

- i) Double Ended Queue (Dequeues)
- ii) Circular Queues

Experiment 7:

Develop a program to generate the binary tree from the given inorder, preorder and post order traversal.

Experiment 8:

Develop a program to implement insertion, deletion and display operation in Min- Max Heap for the given data as integers.

Experiment 9:

Develop a program for Binary Search Tree to implement following operations:

- a) Insertion
- b) Deletion
 - i. Delete node with only child
 - ii. Delete node with both children
- c) Finding an element
- d) Finding Min element
- e) Finding Max element
- f) Left child of the given node
- g) Right child of the given node
- h) Finding the number of nodes, leaves nodes, full nodes, ancestors, descendants

Experiment 10:

Develop a program to implement BFS and DFS for a Graph.

Experiment 11:

Develop a program to implement to generate a min-cost spanning tree

- a) Kruskal's algorithm
- b) Prim's algorithm

Experiment 12:

Develop a program to store k keys into an array of size n at the location computed using a hash function, $loc = key \% n$, where $k \leq n$ and k takes values from [1 to m], $m > n$. To handle the collisions use the following collision resolution techniques,

- a) Linear probing
- b) Quadratic probing
- c) Double hashing/rehashing

Experiment 13:

Develop a program for AVL Tree to implement following operations: (For nodes as integers)

- a) Insertion: Test program for all cases (LL, RR, RL, LR rotation)
- b) Deletion: Test Program for all cases (R0, R1, R-1, L0, L1, L-1)
- c) Display: using set notation

Experiment 14:

Develop a program to generate the B-tree of order 2-3 for a given list of integers.

Experiment 15:

Develop a program to perform string matching using Boyer-Moore algorithm.

Text Books:

1. Data structures, Algorithms and Applications in C++, S.Sahni, 2nd edition, Universities Press, Pvt. Ltd.
2. Data structures and Algorithm Analysis in C++, Mark Allen Weiss, Pearson Education. Ltd., Second Edition.
3. Data structures and Algorithms in C++, Michael T.Goodrich, R.Tamassia and Mount, Wiley student edition, John Wiley and Sons.

Reference Books:

1. Data structures and algorithms in C++, 3rd Edition, Adam Drozdek, Thomson
2. Data structures using C and C++, Langsam, Augenstein and Tanenbaum, PHI.
3. Problem solving with C++, The OOP, Fourth edition, W.Savitch, Pearson education.

Advanced Computing Lab-1

NOTE: Lab programs based on elective taken by student may be offered.

I-I	Advanced Operating Systems Lab	Course Code: V21CTL02	L	T	P	C
			0	0	4	2

Course Outcomes: After completion of course, students would be able to

- CO1:** Demonstrate deadlock avoidance and detection algorithms in a distributed environment (K3)
CO2: Demonstrate efficient clock synchronization and election algorithms. (K3)
CO3: Describe Client server architecture (K2)

List of Experiments

Experiment 1: Develop a program to implement Deadlocks through Simulation.

Experiment 2: Develop a Case Study of 3 tier client server architecture.

Experiment 3: Develop a Case study on Client and RMI Server.

Experiment 4: Develop a program to implement any one Election algorithms.

Experiment 5: Develop a program to show the software simulation for Clock Synchronization in Distributed System using Lamport's Algorithm.

Experiment 6: Develop a program to implement Banker's Algorithm for avoiding Deadlock.

TEXT BOOKS:

1. "Advanced concepts in operating systems: Distributed, Database and multiprocessor operating systems", Mukesh Singhal, Niranjana G.Shivaratri, TMH, 2001.

REFERENCE BOOKS:

1. "Modern operating system", Andrew S.Tanenbaum, PHI, 2003
2. "Distributed operating system-Concepts and design", Pradeep K.Sinha, PHI, 2003.
3. "Distributed operating system", Andrew S.Tanenbaum, Pearson education, 2003.

I SEM	Parallel Computing Lab	Course Code: V21CTL03	L	T	P	C
			0	0	4	2

Course Outcomes: After completion of course, students would be able to

CO1: Develop an efficient parallel algorithm to solve it. (K3)

CO2: Use parallel algorithm using MPI, OpenMP, pthreads, or a combination of MPI and OpenMP.(K3)

List of Experiments

Experiment 1: Develop a parallel program that computes the sum $1+2+\dots+p$ in the following Manner. Each process i assigns the value $i+1$ to an integer, and then the processes perform a sum reduction. As a way of double-checking the result, process 0 should also compute and print the value $p(p+1)/2$.

Experiment 2: Implement A print number is a positive integer evenly divisible by exactly two positive integers: itself and 1. The first five prime numbers are 2,3,5,7 and 11. Sometimes two consecutive odd numbers are both prime. For example, the odd integers following 3,5 and 11 are all prime numbers. However, the odd integer following 7 is not a prime number. Write a parallel program to determine, for all integers less than 1,000,000, the number of times that two consecutive odd integers are both prime.

Experiment 3: Develop a program implementing the parallel matrix multiplication algorithm.

Experiment 4: Develop a MPI program implementing consecutive gradient.

Experiment 5: Develop a MPI program to implementing Quick sort.

Text Books:

1. Parallel Programming in C with MPI and OpenMP Michael J, Quinn Oregon State University.

Reference books:

2. Distributed and Cloud Computing: From Parallel Processing to the Internet of Things 1st Edition, Kai Hwang , Jack Dongarra, Geoffrey C. Fox

I SEM	Advanced Computer Networks Lab	Course Code: V21CTL04	L	T	P	C
			0	0	4	2

Course Outcomes: After completion of course, students would be able to

CO1: Demonstrate various routing protocols. (K3)

CO2: Develop sub netting and addressing IP. (K3)

CO3: Develop emerging trends and security issues in computer Networks. (K3)

List of Experiments

Experiment 1:

Configuration of IP addressing for a given scenario for a given set of topologies.

Experiment 2:

Configure a DHCP Server to serve contiguous IP addresses to a pool of four IP devices with a default gateway and a default DNS address. Integrate the DHCP server with a BOOTP demon to automatically serve Windows and Linux OS Binaries based on client MAC address.

Experiment 3:

Configure, implement and debug the following: Use open source tools for debugging and diagnostics.

- ARP/RARP protocols
- RIP routing protocols
- BGP routing
- OSPF routing protocols
- Static routes (check using netstat)

Text Books:

1. Data communications and networking 4th edition Behrouz A Fourzan, TMH
2. Computer networks 4th edition Andrew S Tanenbaum, Pearson
3. Computer networks, Mayank Dave, CENGAGE

Reference Books:

1. Computer Networks, A system Approach, 5th ed, Larry L Peterson and Bruce S Davie, Elsevier

I SEM	Object Oriented Software Engineering Lab	Course Code: V21CTL05	L	T	P	C
			0	0	4	2

Course Outcomes: After completion of course, students would be able to

CO1: Apply software solutions to problems using an object-oriented strategy. **(K3)**

CO2: Construct the object oriented software systems Model using Unified Modeling Language (UML). **(K3)**

List of Experiments

Experiments 1:

Develop the following for the given project

- Define the problem statement, Software Requirement Specification, entity relationship diagram,
- Dataflow diagrams for level 0 and level 1,
- Draw use-case diagram
- Draw the activity diagram of all use cases.
- Draw sequence diagram of all use cases
- Draw collaboration diagram of all use cases, and Assign objects in Sequence diagram to classes and make class diagrams.

Text Books:

- Object Oriented and Classical Software Engineering, 7/e, Stephen R. Schach, TMH.
- Object oriented and Classical Software Engineering, Timothy Lethbridge, Robert Laganieri, TMH
- Software Engineering by Roger S Pressman, Tata McGraw Hill.

Reference Books:

- Component based Software Engineering: ivica Crnkovic, Springer.

II SEM	Web Technologies	Course Code: V21CTT09	L	T	P	C
			3	0	0	3

Course Outcomes: After completion of course, students would be able to

- CO1:** Demonstrate the basics of JavaScript. (K3)
CO2: Illustrate the concepts of XML and AJAX. (K3)
CO3: Produce Dynamic web pages with PHP and MySQL. (K3)
CO4: Use PERL to retrieve documents from the web. (K3)
CO5: Describe the fundamentals of RUBY Programming. (K2)

Syllabus:

- UNIT-I:** **Java script-** The Basic of Java script: Objects, Primitives Operations and Expressions, Screen Output and Keyboard Input, Control Statements, Object Creation and Modification, Arrays, Functions, Constructors, Pattern Matching using Regular Expressions
- UNIT-II:** **XML-** Document type Definition, XML schemas, Document object model, XSLT, DOM and SAX Approaches, **AJAX A New Approach:** Introduction to AJAX, Integrating PHP and AJAX.
- UNIT-III:** **PHP Programming- Introducing PHP:** Creating PHP script, Running PHP script. **Working with variables and constants:** Using variables, Using constants, Data types, Operators. **Controlling program flow:** Conditional statements, Control statements, Arrays, functions. Working with forms and Databases such as MySQL.
- UNIT-IV:** **PERL-** Introduction to PERL, Operators and if statements, Program design and control structures, Arrays, Hashs and File handling, Regular expressions, Subroutines, Retrieving documents from the web with Perl.
- UNIT-V:** **RUBY:** Introduction to Ruby, Variables, types, simple I/O, Control, Arrays, Hashes, Methods, Classes, Iterators, Pattern Matching. Overview of Rails.

TEXT BOOKS:

1. Programming the World Wide Web, Robert W Sebesta, 7ed, Pearson.
2. Web Technologies, Uttam K Roy, Oxford

Reference Book:

1. The Web Warrior Guide to Web Programming, Bai, Ekedahl, Farrell, Gosselin, Zak, Karparhi, MacIntyre, Morrissey, Cengage.
2. Programming world wide web-Sebesta, Pearson Education, 2007.
3. Core SERVLETS AND JAVA SERVER PAGES VOLUME 1: CORE TECHNOLOGIES By Marty Hall and Larry Brown Pearson Internet and World Wide Web – How to program by Dietel and Nieto PHI/Pearson Education Asia.

II SEM	Data Science through Python Programming	Course Code: V21CTT10	L	T	P	C
			3	0	0	3

Course Outcomes: After completion of course, students would be able to

- CO1:** Illustrate the python basics and the working of various built-in objects in Python. (K2)
CO2: Describe the process of data collection using python. (K2)
CO3: Manipulate data using Pandas and NumPy libraries of python. (K3)
CO4: Apply various techniques for cleaning and pre-processing the data. (K3)
CO5: Demonstrate data visualization techniques using python matplotlib. (K3)

Syllabus:

- UNIT I: PYTHON Basics and Programming Concepts:** Introducing Python, Types and Operations - Numbers, Strings, Lists, Tuples, Dictionaries, Files, Numeric Types, Dynamic Typing; Statements and Syntax - Assignments, Expressions, Statements, Loops, iterations, comprehensions; Functions - Function Basics, Scopes, Arguments, Advanced Functions; Modules - Module Coding Basics, Module Packages, Advanced Module Topics; Classes and OOP - Class, Operator Overloading, Class Designing; Exceptions and Tools - Exception Basics, Exception Coding Details, Exception Objects, Designing With Exceptions, Parallel System Tools
- UNIT II: GUI Programming:** Graphical User Interface - Python gui development options, Adding Widgets, GUI Coding Techniques, Customizing Widgets; Internet Programming - Network Scripting, Client-Side scripting, Pymailgui client, server-side scripting, Pymailcgi server; Tools and Techniques - databases and persistence, data structures, text and language, python/c integration
- UNIT III: Pandas and NumPy:** Numpy Basics - Fast Element wise array functions, Multidimensional Array, Data Processing using arrays, file i/o with arrays; Pandas - Data Structures, Essential Functionality, Summarizing and Computing Descriptive Statistics, Handling Missing Data, Hierarchical Indexing
- UNIT IV: Data Preprocessing:** Data Loading, Storage, and FileFormats - Reading and Writing data in text format, binary data formats, interacting with html and web apis, interacting with databases; Data Wrangling: Clean, Transform, Merge, Reshape - Combining and Merging Data Sets, Reshaping and Pivoting, Data Transformation, String Manipulation; Data Aggregation and Group Operations – Group by Mechanics, Data Aggregation, Groupby Operations and and Transformations, Pivot Tables and Cross- Tabulation
- UNIT V: Data Visualization:** A Brief matplotlib API Primer, Plotting Functions in pandas, Time Series, Financial and Economic Data Applications

Text Books:

1. Learning Python , O'Reilly, Mark Lutz
2. Programming Python, O'Reilly, Mark Lutz
3. Python For Data Analysis (O'Reilly, Wes Mckinney)

Reference Text Books:

1. Python: The Complete Reference, Martin C. Brown, McGraw Hill Education
2. Head First Python, Paul Barry, O'Reilly

II SEM	Machine Learning (Elective –III)	Course Code: V21CTT11	L	T	P	C
			3	0	0	3

Course Outcomes: After completion of course, students would be able to

- CO1:** Describe Knowledge for Productive use of Machine Learning and Diversity of Data (K2)
CO2: Demonstrate on Supervised and Computational Learning (K3)
CO3: Illustrate on Statistics in learning techniques and Logistic Regression (K3)
CO4: Illustrate on Support Vector Machines and Perceptron Algorithm (K3)
CO5: Construct a Multilayer Perceptron Networks and classification of decision tree (K3)

UNIT-I: Introduction-Towards Intelligent Machines, Well posed Problems, Example of Applications in diverse fields, Data Representation, Domain Knowledge for Productive use of Machine Learning, Diversity of Data: Structured / Unstructured, Forms of Learning, Machine Learning and Data Mining, Basic Linear Algebra in Machine Learning Techniques.

UNIT-II: Supervised Learning- Rationale and Basics: Learning from Observations, Bias and Why Learning Works: Computational Learning Theory, Occam's Razor Principle and Overfitting Avoidance Heuristic Search in inductive Learning, Estimating Generalization Errors, Metrics for assessing regression, Metrics for assessing classification.

UNIT-III: Statistical Learning- Machine Learning and Inferential Statistical Analysis, Descriptive Statistics in learning techniques, Bayesian Reasoning: A probabilistic approach to inference, K-Nearest Neighbor Classifier. Discriminant functions and regression functions, Linear Regression with Least Square Error Criterion, Logistic Regression for Classification Tasks, Fisher's Linear Discriminant and Thresholding for Classification, Minimum Description Length Principle.

UNIT-IV: Support Vector Machines (SVM)- Introduction, Linear Discriminant Functions for Binary Classification, Perceptron Algorithm, Large Margin Classifier for linearly separable data, Linear Soft Margin Classifier for Overlapping Classes, Kernel Induced Feature Spaces, Nonlinear Classifier, Regression by Support vector Machines.

Learning with Neural Networks: Towards Cognitive Machine, Neuron Models, Network Architectures, Perceptrons, Linear neuron and the Widrow-Hoff Learning Rule, The error correction delta rule.

UNIT -V: Multilayer Perceptron Networks and error back propagation algorithm, Radial Basis Functions Networks. **Decision Tree Learning:** Introduction, Example of classification decision tree, measures of impurity for evaluating splits in decision trees, ID3, C4.5, and CART decision trees, pruning the tree, strengths and weakness of decision tree approach.

Textbooks:

1. Applied Machine Learning, 1e, M.Gopal, Mc Graw Hill Education, 2018

Reference Books

1. Kevin Murphy, Machine Learning: A Probabilistic Perspective, MIT Press, 2012
2. Trevor Hastie, Robert Tibshirani, Jerome Friedman, The Elements of Statistical Learning, Springer 2009 (freely available online)
3. Christopher Bishop, Pattern Recognition and Machine Learning, Springer, 2007.

II SEM	Ad Hoc & Sensor Networks (Elective –III)	Course Code: V21CTT12	L	T	P	C
			3	0	0	3

Course Outcomes: After completion of course, students would be able to

- CO1:** Explain the Fundamental Concepts and applications of ad hoc and wireless sensor networks (K2)
CO2 : Describe the MAC protocol issues of ad hoc networks (K2)
CO3 : Describe routing protocols for ad hoc wireless networks with respect to TCP design issues (K2)
CO4 : Explain the concepts of network architecture and MAC layer protocol for WSN (K2)
CO5 : Discuss the WSN routing issues by considering QoS measurements (K2)

UNIT I: Introduction : Fundamentals of Wireless Communication Technology, The Electromagnetic Spectrum, Radio propagation Mechanisms ,Characteristics of the Wireless channel mobile ad hoc networks (MANETs), **Wireless Sensor Networks (WSNs):** concepts and architectures, Applications of Ad Hoc and Sensor Networks, Design Challenges in Ad hoc and Sensor Networks.

UNIT II: MAC Protocols For Ad Hoc Wireless Networks: Issues in designing a MAC Protocol, Issues in Designing a MAC Protocol for Ad Hoc Wireless Networks, Design Goals of a MAC Protocol for Ad Hoc Wireless Networks, Classification of MAC Protocols, Contention based protocols, Contention based protocols with Reservation Mechanisms, Contention based protocols with Scheduling Mechanisms, Multi channel MAC - IEEE 802.11.

UNIT III: Routing Protocols And Transport Layer In Ad Hoc Wireless Networks: Routing Protocol: Issues in designing a routing protocol for Ad hoc networks, Classification, proactive routing, reactive routing (on-demand), hybrid routing, Transport Layer protocol for Ad hoc networks, Design Goals of a Transport Layer Protocol for Ad Hoc Wireless Networks, Classification of Transport Layer solutions- TCP over Ad hoc wireless, Network Security, Security in Ad Hoc Wireless Networks, Network Security Requirements.

UNIT IV: Wireless Sensor Networks (WSNs) And Mac Protocols: Single node architecture - hardware and software components of a sensor node, **WSN Network architecture:** typical network architectures, data relaying and aggregation strategies, **MAC layer protocols:** self-organizing, Hybrid TDMA/FDMA and CSMA based MAC -IEEE 802.15.4.

UNIT V: WSN Routing, Localization & Qos: Issues in WSN routing, OLSR, Localization, Indoor and Sensor Network Localization, absolute and relative localization, triangulation, QOS in WSN, Energy Efficient Design, Synchronization.

Text Books:

1. "Ad Hoc Wireless Networks: Architectures and Protocols ", C. Siva Ram Murthy, and B. S. Manoj, Pearson Education, 2008
2. "Wireless Adhoc and Sensor Networks", Labiod. H, Wiley, 1st edition-2008
3. "Wireless ad -hoc and sensor Networks: theory and applications", Li, X, Cambridge University Press, fifth edition-2008.

Reference Books:

1. "Ad Hoc & Sensor Networks: Theory and Applications", 2nd edition, Carlos De MoraesCordeiro, Dharma Prakash Agrawal ,World Scientific Publishing Company, 2011
2. "Wireless Sensor Networks", Feng Zhao and LeonidesGuibas,Elsevier Publication 2nd edition-2004
3. "Protocols and Architectures for Wireless Sensor Networks", Holger Karl and Andreas Willig,Wiley, 2005 (soft copy available)
4. "Wireless Sensor Networks Technology, Protocols, and Applications", KazemSohraby, Daniel Minoli, &TaiebZnati, John Wiley, 2007. (soft copy available)

II SEM	Internet of Things (Elective –III)	Course Code: V21CTT13	L	T	P	C
			3	0	0	3

Course Outcomes: After completion of course, students would be able to

- CO1:** Describe the term 'internet of things' in different contexts. (K2)
CO2: Develop various protocols for IoT. (K3)
CO3: Develop a PoC of an IoT system using Raspberry Pi/Arduino (K3)
CO4: Apply data analytics and use cloud offerings related to IoT. (K3)
CO5: Demonstrate applications of IoT in real time scenario (K3)

UNIT I: FUNDAMENTALS OF IoT: Evolution of Internet of Things, Enabling Technologies, IoT Architectures, oneM2M, IoT World Forum (IoTWF) and Alternative IoT models, Simplified IoT Architecture and Core IoT Functional Stack, Fog, Edge and Cloud in IoT, Functional blocks of an IoT ecosystem, Sensors, Actuators, Smart Objects and Connecting Smart Objects

UNIT II: IoT PROTOCOLS: IT Access Technologies: Physical and MAC layers, topology and Security of IEEE 802.15.4, 802.15.4g, 802.15.4e, 1901.2a, 802.11ah and Lora WAN, Network Layer: IP versions, Constrained Nodes and Constrained Networks, Optimizing IP for IoT: From 6LoWPAN to 6Lo, Routing over Low Power and Lossy Networks, Application Transport Methods: Supervisory Control and Data Acquisition, Application Layer Protocols: CoAP and MQTT. Bluetooth Smart Connectivity-Overview, Key Versions, BLE-Bluetooth Low Energy Protocol, Low Energy Architecture.

UNIT III: DESIGN AND DEVELOPMENT: Design Methodology, Embedded computing logic, Microcontroller, System on Chips, IoT system building blocks, Arduino, Board details, IDE programming, Raspberry Pi, Interfaces and Raspberry Pi with Python Programming.

UNIT IV: Arm Based Embedded System Design: ARM Cortex-A class processor, Embedded Devices-ARM Cortex-M Class processor, Networking-Bluetooth Smart Technology
Introduction to embedded systems: CPUs vs MCU's vs Embedded Systems, Examples, Options for Building Embedded Systems, Features of Embedded Systems, Building Embedded Systems, Building Embedded Systems using MCUs, Introduction to mbedTM Platform

UNIT V: CASE STUDIES/INDUSTRIAL APPLICATIONS: Cisco IoT system, IBM Watson IoT platform, Manufacturing, Converged Plant wide Ethernet Model (CPwE), Power Utility Industry, Grid Blocks Reference Model, Smart and Connected Cities: Layered architecture, Smart Lighting, Smart Parking Architecture and Smart Traffic Control.

Text Books:

1. IoT Fundamentals: Networking Technologies, Protocols and Use Cases for Internet of Things, David Hanes, Gonzalo Salgueiro, Patrick Grossetete, Rob Barton and Jerome Henry, Cisco Press, 2017
2. The Definitive Guide to ARM Cortex-M3 and M4 Processors, 3rd Edition, Joseph Yiu

Reference Books:

1. Internet of Things – A hands-on approach, Arshdeep Bahga, Vijay Madisetti, Universities Press, 2015
2. The Internet of Things – Key applications and Protocols, Olivier Hersent, David Boswarthick, Omar Elloumi and Wiley, 2012 (for Unit 2).
3. "From Machine-to-Machine to the Internet of Things – Introduction to a New Age of Intelligence", Jan Ho"ller, Vlasios Tsiatsis, Catherine Mulligan, Stamatis, Karnouskos, Stefan Avesand, David Boyle and Elsevier, 2014.
4. Architecting the Internet of Things, Dieter Uckelmann, Mark Harrison, Michahelles and Florian (Eds), Springer, 2011.
5. Recipes to Begin, Expand, and Enhance Your Projects, 2nd Edition, Michael Margolis, Arduino Cookbook and O'Reilly Media, 2011.
6. Cortex–A series Programmer's Guide for ARMv7-A by Arm
<http://infocenter.arm.com/help/index.jsp?topic=/com.arm.doc.den0013d/index.htmlc>

II SEM	Principles of Cyber Security (Elective-IV)	Course Code: V21CTT14	L	T	P	C
			3	0	0	3

Course Outcomes: After completion of course, students would be able to

- CO1:** Apply cyber security architecture principles. (K3)
- CO2:** Describe risk management processes and practices (K2).
- CO3:** Construct cyber security incidents to apply appropriate response (K3)
- CO4:** Differentiate system and application security threats and vulnerabilities. (K2)
- CO5:** Identify security tools and hardening techniques. (K1)

- UNIT-I: Introduction to Cyber security-** Cyber security objectives, Cyber security roles, Differences between Information Security & Cyber security, **Cyber security Principles-** Confidentiality, integrity, & availability Authentication & non- repudiation.
- UNIT-II: Information Security (IS) within Lifecycle Management-** Lifecycle management landscape, Security architecture processes, Security architecture tools, Intermediate lifecycle management concepts, **Risks & Vulnerabilities-** Basics of risk management, Operational threat environments, Classes of attacks.
- UNIT-III: Incident Response-** Incident categories, Incident response Incident recovery, and **Operational security protection:** Digital and data assets, ports and protocols, Protection technologies, Identity and access Management, configuration management.
- UNIT-IV: Threat Detection and Evaluation (DE): Monitoring-** Vulnerability Management, Security Logs and Alerts, Monitoring Tools and Appliances. **Analysis-** Network traffic Analysis, packet capture and analysis .
- UNIT-V: Introduction to backdoor System and security-** Introduction to metasploit, Backdoor, demilitarized zone(DMZ), Digital Signature, Brief study on Hardening of operating system.

Text Books:

1. NASSCOM: Security Analyst Student Hand Book Dec 2015.
2. Information Security Management Principles Updated Edition by David Alexander, Amanda Finch, David Sutton ,Published by BCS, June 2013.

Reference Books:

1. CSX- cyber security fundamentals 2nd edition, Published by ISACA, Cyber security, Network Security, Data Governance Security.

II SEM	Cloud Computing (Elective –IV)	Course Code: V21CTT15	L	T	P	C
			3	0	0	3

Course Outcomes: After completion of course, students would be able to

- CO1:** Interpret the key dimensions of the challenge of Cloud Computing (K2)
CO2: Examine the economics, financial and technological implications for selecting cloud computing for own organization. (K3)
CO3: Explain the financial, technological, and organizational capacity of employer's for actively initiating and installing cloud-based applications (K2)
CO4: Examine the own organizations' needs for capacity building and training in cloud computing-related IT areas. (K3)
CO5: Illustrate Virtualization for Data-Center Automation. (K2)

SYLLABUS:

- UNIT I: Introduction:** Network centric computing, Network centric content, peer-to –peer systems, cloud computing delivery models and services, Ethical issues, Vulnerabilities, Major challenges for cloud computing. **Parallel and Distributed Systems:** Introduction, architecture, distributed systems, communication protocols, logical clocks, message delivery rules, concurrency, model concurrency with Petri Nets.
- UNIT II: Cloud Infrastructure:** At Amazon, The Google Perspective, Microsoft Windows Azure, Open Source Software Platforms, Cloud storage diversity, Inter cloud, energy use and ecological impact, responsibility sharing, user experience, Software licensing, **Cloud Computing:** Applications and Paradigms: Challenges for cloud, existing cloud applications and new opportunities, architectural styles, workflows, The Zookeeper, The Map Reduce Program model, HPC on cloud, biological research.
- UNIT III: Cloud Resource virtualization:** Virtualization, layering and virtualization, virtual machine monitors, virtual machines, virtualization- full and para, performance and security isolation, hardware support for virtualization, Case Study: Xen, vBlades, **Cloud Resource Management and Scheduling:** Policies and Mechanisms, Applications of control theory to task scheduling, Stability of a two-level resource allocation architecture, feedback control based on dynamic thresholds, coordination, resource bundling, scheduling algorithms, fair queuing, start time fair queuing, cloud scheduling subject to deadlines, Scheduling Map Reduce applications, Resource management and dynamic application scaling.
- UNIT IV: Storage Systems:** Evolution of storage technology, storage models, file systems and database, distributed file systems, general parallel file systems. Google file system. Apache Hadoop, Big Table, Megastore (text book 1), Amazon Simple Storage Service(S3) (Text book 2), **Cloud Security:** Cloud security risks, security – a top concern for cloud users, privacy and privacy impact assessment, trust, OS security, Virtual machine security, Security risks.
- UNIT V: Cloud Application Development:** Amazon Web Services : EC2 – instances, connecting clients, security rules, launching, usage of S3 in Java, Installing Simple Notification Service on Ubuntu 10.04, Installing Hadoop on Eclipse, Cloud based simulation of a Distributed trust algorithm, Cloud service for adaptive data streaming (Text Book 1), **Google:** Google App Engine, Google Web Toolkit (Text Book 2), **Microsoft:** Azure Services Platform, Windows live, Exchange Online, Share Point Services, Microsoft Dynamics CRM (Text Book 2).

Text Books:

1. Cloud Computing, Theory and Practice, Dan C Marinescu, MK Elsevier
2. Cloud Computing, A Practical Approach, Anthony T Velte, Toby J Velte, Robert Elsenpeter, TMH

Reference book:

Mastering Cloud Computing, Foundations and Application Programming, Raj Kumar Buyya, Christen vecctiola, S Tammarai selvi, TMH

II SEM	Natural Language Processing (Elective –IV)	Course Code: V21CTT16	L	T	P	C
			3	0	0	3

Course Outcomes: After completion of course, students would be able to

- CO1:** Explain approaches to syntax and semantics in NLP. (K2)
- CO2:** Demonstrate approaches to discourse, generation, dialogue and summarization within NLP. (K3)
- CO3:** Explain current methods for statistical approaches to machine translation. (K2)
- CO4:** Identify machine learning techniques used in NLP, including hidden Markov models and probabilistic (K2)
- CO5:** Explain context-free grammars, clustering and unsupervised methods, log-linear and discriminative models, and the EM algorithm as applied within NLP (K2)

UNIT-I: **Introduction:** NLP tasks in syntax, semantics, and pragmatics. Applications such as information extraction, question answering, and machine translation. The problem of ambiguity. The role of machine learning. Brief history of the field.

UNIT-II: **N-gram Language Models:** The role of language models, Simple Ngram models. Estimating parameters and smoothing. Evaluating language models. **Part of Speech Tagging and Sequence Labeling:** Lexical syntax. Hidden Markov Models. Maximum Entropy Models. Conditional Random Fields

UNIT-III: **Syntactic parsing:** Grammar formalisms and tree banks. Efficient parsing for context-free grammars (CFGs). Statistical parsing and probabilistic CFGs (PCFGs). Lexicalized PCFGs.

UNIT-IV: **Semantic Analysis:** Lexical semantics and word-sense disambiguation. Compositional semantics. Semantic Role Labeling and Semantic Parsing.

UNIT- V: **Information Extraction (IE) and Machine Translation (MT):**
Named entity recognition and relation extraction. IE using sequence labeling. Basic issues in MT. Statistical translation, word alignment, phrase based translation, and synchronous grammars. Dialogues: Turns and utterances, grounding, dialogue acts and structures Natural Language Generation: Introduction to language generation, architecture, discourse planning (text schemata, rhetorical relations).

Text Books:

1. D. Jurafsky & J. H. Martin – “Speech and Language Processing – An introduction to Language processing, Computational Linguistics, and Speech Recognition”, Pearson Education

References:

1. Allen, James. 1995. – “Natural Language Understanding”. Benjamin/ Cummings, 2ed.
2. Bharathi, A., Vineet Chaitanya and Rajeev Sangal. 1995. Natural Language Processing- “A Pananian Perspective”. Prentice Hall India, Eastern Economy Edition.
3. Eugene Charniak: “Statistical Language Learning”, MIT Press, 1993.
4. Manning, Christopher and Heinrich Schutze. 1999. “Foundations of Statistical Natural Language Processing”. MIT Press.

II SEM	Advanced Web Technologies Lab	Course Code: V21CTL06	L	T	P	C
			0	0	4	2

Course Outcomes: After completion of course, students would be able to

CO1:	Develop static web pages using HTML, CSS.	(K3)
CO2:	Demonstrate the concepts of JavaScript and DHTML.	(K2)
CO3:	Demonstrate the basic concepts of PHP and JSP.	(K2)
CO4:	Demonstrate the concepts of Extensible markup language & AJAX.	(K2)
CO5:	Develop dynamic Web Applications using PHP & MySQL.	(K3)

List of Experiments

1.
 - a) Create a Simple HTML home page provides links to move to other pages like hobbies, educational info, personal info etc.
 - b) Create a HTML program to illustrate the use of frame and frameset tags of HTML.
 - c) Create a HTML Program which use a HTML controls to create a student information form to collect student's information like name, address, phone, email, sex, birth date, hobbies etc.
2.
 - a) Create a webpage which displays "Hello World" with font size 20 pixels, bold format, in "Times New Roman" font and green in color using inline CSS, embedded CSS and external CSS.
 - b) Create a webpage which displays the class time table and apply the following effects on the table:
 - For the table header apply *blue* as the background color and *white* for the color of the text in the table header.
 - Display *day names* (Mon, Tue etc...) in bold format with the first letter in the day name in uppercase.
 - ☐ Display *lunch* slightly in bigger font other than the remaining text.
 - c) Create a webpage to manage personal details like name, class, qualifications, photo, address etc., using tables and other suitable HTML tags. Apply the following style information:
 - Display the heading of the page in *Times New Roman* font and with 24px size.
 - Align all the field names like Name, Class, Photo etc to *right* in the table.
 - Apply *light blue* as background color for the left side cells in the table which contains field names like Name, Class etc...
 - Also display your college logo as background image in the top right position of the web page.
 - d) Create a web page containing two images, where one image overlaps another image by using the *z-index* CSS property.
3.
 - a) Create a HTML Program which demonstrates loops like for loop, do while, while in java script.
 - b) Create a HTML Program which demonstrates the use of functions in java script.
 - c) Create a HTML Program which demonstrates various events like onclick, ondblclick, onfocus, onblur, onchange, onmouseover, onmouseover, window event, onload, onunload event.
 - d) Create a HTML Program to create various functions and sub routines to validate the data entered by user in form.
4.
 - a) Create a program to illustrate the concept of associative array in PHP.
 - b) Create PHP program to implement the concept of Session management.
 - c) Create a PHP program to display student information in webpage. Student's data is stored in My SQL database.
 - d) Create a PHP program to insert student information from HTML form. Student's data is

- stored in My SQL database.
5.
 - a) Create a well-formed XML document.
 - b) Create a valid XML document using DTD.
 - c) Create a valid XML document using XML Schema.
 - d) Create a XML document which contains details of cars and display the same as a table using XSLT.
 - e) Write a Java program to parse the XML document containing car details using SAX API.
 6.
 - a) Create a servlet to display "Hello World" in the browser.
 - b) Create a servlet to store email-id as an initialization parameter and print the same email-id by reading the initialization parameter from the web.xml file.
 - c) Create a servlet to retrieve name and branch details from a html page and print the same using the servlet.
 - e) Create a HTML page which accepts book id, book name and book price and a submit button. When the user clicks the submit button, all the values assigned to the previous text fields must be stored in a session object and the control forwards to another servlet where the values stored in the session are retrieved and displayed.
 7.
 - a) Create a JSP page to display "Hello World" in the browser.
 - b) Create a JSP page to store email-id as an initialization parameter and print the same email-id by reading the initialization parameter from the web.xml file.
 - c) Create a JSP page to retrieve name and branch details from a html page and print the same using a servlet.
 - d) Create a HTML page which accepts book id, book name and book price and a submit button. When the user clicks the submit button, all the values assigned to the previous text fields must be stored in a session object and the control forwards to a JSP page where the values stored in the session are retrieved and displayed.
 8. Create a HTML page which accepts student regd.no. and prints the results of that student by retrieving the results from the database. Use AJAX to display the "please wait..." while the server is processing the request and print the result of the student when the server returns the result. Server resource can be either servlet or JSP or PHP

Reference Books:

1. *Java server programming java JavaEE5 Black Book*l, Kogent Solutions Dreamtech Press, Inc, ISBN-13 9788177228359 ISBN-10 8177228358, 2008.
2. *AJAX black book*l, new edition, Kogent Solutions Inc, Dreamtech Press, ISBN:10- 81-7722-838-2 ISBN:13-978-81-7722-838-063. Jonathan Chaffer, Karl Swedberg, —*Learning jQuery*l, 3rd Edition , , ISBN 13: 9781849516549, 2011
3. Chris Bates, *Web Programming- building internet applications*, 2nd edition, WILEY Dreamtech, 2006
4. Patrick Naughton and Herbert Schildt, *The complete Reference Java seventh Edition*, TMH, 2007
5. Hans Bergsten, *Java Server Pages*, SPD O'Reilly, 2000
6. *Java Server Programming*, Ivan Bayross and others, The X Team, SPD
7. *Web Warrior Guide to Web Programmimg*-Bai/Ekedaw-Thomas
8. *Beginning Web Programming*-Jon Duckett WROX.
9. *Java Server Pages*, Pekowsky, Pearson.
10. *Java Script*, D.Flanagan, O'Reilly, SPD.

II SEM	Data Science Applications with Python Lab	Course Code: V21CTL07	L	T	P	C
			0	0	4	2

Course Outcomes: After completion of course, students would be able to

CO1: Use data science operations like data collection, management and storing. (K3)

CO2: Apply Python programming concepts in data science, including their real-world applications. (K3)

CO3: Develop data collection and management scripts using Python Pandas. (K3)

List of Experiments

Experiment 1:

Develop a Python Program to Find the Sum of the Series: $1 + 1/2 + 1/3 + \dots + 1/N$

Experiment 2:

Develop a Python Program to Split the array and add the first part to the end

Experiment 3:

Develop a Python Program to Create a List of Tuples with the First Element as the Number and Second Element as the Square of the Number

Experiment 4:

Develop a Python program to count number of vowels using sets in given string

Experiment 5:

Develop a program to implement permutation of a given string using inbuilt function.

Experiment 6:

Develop a python program to sort list of dictionaries by values in Python – Using lambda function.

Experiment 7:

Develop a Python Program for following sorting:
i. Quick Sort ii. Heap Sort

Experiment 8:

Develop a Python Program to Reverse a String Using Recursion

Experiment 9:

Develop a Python Program to Count the Number of Words in a Text File

Experiment 10:

Develop a Python Program to Read the Contents of a File in Reverse Order

Experiment 11:

Develop a program to Merge and Join DataFrames with Pandas in Python

Experiment 12:

Develop a program to implement Merge and Join Data Frames with Python Pandas

Experiment 13:

Develop a Python Program to Append the Contents of One File to Another File

Experiment 14:

Explain how to install and Load CSV files to Python Pandas

Text Books:

1. Learning Python , O'Reilly, Mark Lutz
2. Programming Python, O'Reilly, Mark Lutz
3. Python For Data Analysis (O'Reilly, Wes Mckinney)

III SEM	Mobile Applications and Development (Elective-V)	Course Code: V21CTT18	L	T	P	C
			3	0	0	3

Course Outcomes: After completion of course, students would be able to

- CO1:** Describe Installation and configuration of Android application development tools. (K2)
CO2: Develop applications using services and publishing android applications. (K3)
CO3: Demonstrate Android software development tools. (K3)
CO4: Illustrate debugging programs running on mobile devices. (K2)
CO5: Develop Android applications using server-less database like SQLite. (K3)

Unit I: **Introduction to Android:** The Android 4.1 jelly Bean SDK, Understanding the Android Software Stack, installing the Android SDK, Creating Android Virtual Devices, Creating the First Android Project, Using the Text view Control, Using the Android Emulator, The Android Debug Bridge(ADB), Launching Android Applications on a Handset.

Unit II: **Basic Widgets:** Understanding the Role of Android Application Components, Understanding the Utility of Android API, Overview of the Android Project Files, Understanding Activities, Role of the Android Manifest File, Creating the User Interface, Commonly Used Layouts and Controls, Event Handling, Displaying Messages Through Toast, Creating and Starting an Activity, Using the Edit Text Control, Choosing Options with Checkbox, Choosing Mutually Exclusive Items Using Radio Buttons.

Unit III: **Building Blocks for Android Application Design:** Introduction to Layouts, Linear Layout, Relative Layout, Absolute Layout, Using Image View, Frame Layout, Table Layout, Grid Layout, Adapting to Screen orientation. Utilizing Resources and Media Resources, Creating Values Resources, Using Drawable Resources, Switching States with Toggle Buttons, Creating an Images Switcher Application, Scrolling Through Scroll View, playing Audio, Playing Video, Displaying Progress with Progress Bar, Using Assets.

Unit IV: **Using Selection widgets and Debugging:** Using List View, Using the Spinner control, Using the GridView Control, Creating an Image Gallery Using the ViewPager Control, Using the Debugging Tool: Dalvik Debug Monitor Service(DDMS), Debugging Application, Using the Debug Perspective. Displaying And Fetching Information Using Dialogs and Fragments: What Are Dialogs?, Selecting the Date and Time in One Application, Fragments, Creating Fragments with java Code, Creating Special Fragments

Unit V: **Building Menus and Storing Data:** Creating Interface Menus and Action Bars, Menus and Their Types, Creating Menus Through XML, Creating Menus Through Coding, Applying a Context Menu to a List View, Using the Action Bar, Replacing a Menu with the Action Bar, Creating a Tabbed Action Bar, Creating a Drop-Down List Action Bar Using Databases: Using the SQLiteOpenHelperclasss, Accessing Databases with the ADB, Creating a Data Entry Form, Communicating with SMS and Emails: Understanding Broadcast Receivers, Using the Notification System, Sending SMS Messages with Java Code, Receiving SMS Messages, Sending Email, Working With Telephony Manager.

Text Books

1. Android Programming by B.M Harwani, Pearson Education, 2013.

References Text Books:

1. Android application Development for Java Programmers, James C Sheusi, Cengage Learning
2. Android In Action by w.Frank Ableson, Robi Sen, Chris King, C. Enrique Ortiz., Dreamtech.
3. Professional Android 4 applications development, Reto Meier, Wiley India, 2012.
4. Beginning Android 4 applications development, Wei- Meng Lee, Wiley India, 2013.

III SEM	Big Data Analytics (Elective-V)	Course Code: V21CTT19	L	T	P	C
			3	0	0	3

Course Outcomes: After completion of course, students would be able to

- CO1:** Illustrate big data and its use cases from selected business domains. (K3)
CO2: Interpret and summarize No SQL and Cassandra (K3)
CO3: Discuss the HADOOP and Map Reduce technologies associated with big data analytics and explore on Big Data applications Using Hive. (K2)
CO4: Define the use of Apache Spark, RDDs etc. to work with datasets. (K1)
CO5: Assess real time processing with Spark Streaming. (K3)

Syllabus:

- UNIT I:** What is big data, why big data, convergence of key trends, unstructured data, industry examples of big data, web analytics, big data and marketing, fraud and big data, risk and big data, credit risk management, big data and algorithmic trading, big data and healthcare, big data in medicine, advertising and big data, big data technologies, introduction to Hadoop, open source technologies, cloud and big data, mobile business intelligence, Crowd sourcing analytics, inter and trans firewall analytics.
- UNIT II:** Introduction to NoSQL, aggregate data models, aggregates, key-value and document data models, relationships, graph databases, schema less databases, materialized views, distribution models, sharding, master-slave replication, peer- peer replication, sharding and replication, consistency, relaxing consistency, version stamps, Working with Cassandra ,Table creation, loading and reading data.
- UNIT III:** Data formats, analyzing data with Hadoop, scaling out, Architecture of Hadoop distributed file system (HDFS), fault tolerance ,with data replication, High availability, Data locality , Map Reduce Architecture, Process flow, Java interface, data flow, Hadoop I/O, data integrity, compression, serialization. Introduction to Hive, data types and file formats, HiveQL data definition, HiveQL data manipulation, Logical joins, Window functions, Optimization, Table partitioning, Bucketing, Indexing, Join strategies.
- UNIT IV:** Apache spark- Advantages over Hadoop, lazy evaluation, In memory processing, DAG, Spark context, Spark Session, RDD, Transformations- Narrow and Wide, Actions, Data frames ,RDD to Data frames, Catalyst optimizer, Data Frame Transformations, Working with Dates and Timestamps, Working with Nulls in Data, Working with Complex Types, Working with JSON, Grouping, Window Functions, Joins, Data Sources, Broadcast Variables, Accumulators, Deploying Spark- On-Premises Cluster Deployments, Cluster Managers- Standalone Mode, Spark on YARN , Spark Logs, The Spark UI- Spark UI History Server, Debugging and Spark First Aid
- UNIT V:** Spark-Performance Tuning, Stream Processing Fundamentals, Event-Time and State full Processing - Event Time, State full Processing, Windows on Event Time- Tumbling Windows, Handling Late Data with Watermarks, Dropping Duplicates in a Stream, Structured Streaming Basics - Core Concepts, Structured Streaming in Action, Transformations on Streams, Input and Output.

Text Books:

1. Big Data, Big Analytics: Emerging, Michael Minnelli, Michelle Chambers, and Ambiga Dhiraj
2. SPARK: The Definitive Guide, Bill Chambers & Matei Zaharia, O'Reilly, 2018 Edition
3. Business Intelligence and Analytic Trends for Today's Businesses", Wiley, 2013
4. P. J. Sadalage and M. Fowler, "NoSQL Distilled: A Brief Guide to the Emerging World Polyglot Persistence", Addison-Wesley Professional, 2012
5. Tom White, "Hadoop: The Definitive Guide", Third Edition, O'Reilly, 2012

Reference Books:

1. "Hadoop Operations", O'Reilly, Eric Sammer, 2012
2. "Programming Hive", O'Reilly, E. Capriolo, D. Wampler, and J. Rutherglen, 2012
3. "HBase: The Definitive Guide", O'Reilly, Lars George, 2011
4. "Cassandra: The Definitive Guide", O'Reilly, Eben Hewitt, 2010
5. "Programming Pig", O'Reilly, Alan Gates, 2011.

III SEM	Operations Research (Open Elective)	Course Code: V21MBT56	L	T	P	C
			3	0	0	3

Course Outcomes: After completion of course, students would be able to

CO 1: solve the linear programming problem.

CO 2: solve Transportation and Assignment problems.

CO 3: apply sequencing techniques to create the jobs.

CO 4: solve problems of waiting lines.

CO 5: apply the principles of Game theory to real world competitive situations.

UNIT-I: Introduction to Operations Research: Basics definition, scope, objectives, phases, models and limitations of Operations Research. Linear Programming Problem – Formulation of LPP, Graphical solution of LPP. Simplex Method, Artificial variables, big-M method, degeneracy and unbound solutions.

UNIT-II: Transportation Problem. Formulation, solution, unbalanced Transportation problem. Finding basic feasible solutions – Northwest corner rule, least cost method and Vogel's approximation method. Optimality test: the stepping stone method and MODI method.

UNIT-III: Assignment model. Formulation. Hungarian method for optimal solution. Solving unbalanced problem. Traveling salesman problem and assignment problem Sequencing models. Solution of Sequencing Problem – Processing n Jobs through 2 Machines – Processing n Jobs through 3 Machines – Processing 2 Jobs through m machines – Processing n Jobs through m Machines.

UNIT-IV: Queuing Theory: Introduction – Single Channel – Poisson arrivals – exponential service times – with infinite population and finite population models – multi channel – poisson arrivals – exponential service times with infinite population single channel poisson arrivals

UNIT-V: Game Theory: Introduction – Minimax and Maximin – Criterion and optimal strategy – Solution of games with saddle points – Rectangular games without saddle points – algebraic method – Graphical Method (m x 2, 2 x n methods) – dominance principle .

Text Books:

1. Operations Research by J.K.Sharma; Publisher: Mac Milan.
2. A. M. Natarajan, P.Balasubramani, A.Tamilarasi, Operations Research by Publisher: Pearson Education.

References:

1. Operations Research by R.Pannerselvam; Publisher: Prentice Hall international.
2. P.Sankar Iyer, "Operations Research" Tata McGraw- Hill, 2008.
3. S.D.Sharma Operations Research; Kedarnath
4. Wayne L. Winston, Jeffrey B.Golbary, Operations Research Thomson / Brooks/Cole Publication.

III SEM	Cost Management of Engineering Projects(Open Elective)	Course Code: V21MBT56	L	T	P	C
			3	0	0	3

Course Outcomes: After completion of course, students would be able to

CO1: Understand the cost management process and various costs involved in a project (K2)

CO2: Understand various aspects of a project and related processes. (K2)

CO3: Analyze the concepts of Break even and CVP analysis. (K3)

CO4: Demonstrate quality management techniques besides budgeting strategies (K2)

CO5: Apply quantitative techniques for cost management (K4)

UNIT I: Introduction and Overview of the Strategic Cost Management Process: Cost concepts in decision-making; relevant cost, Differential cost, Marginal cost, Incremental cost and Opportunity cost. Objectives of Costing System; Creation of a Database for operational control; Provision of data for Decision-Making.

UNIT II: Project Management: Meaning, Different types of projects. **Various stages of project execution:** conception to commissioning, Project execution as a conglomeration of technical and nontechnical activities, Detailed Engineering activities, Pre project execution, main clearances and documents. **Project team:** Role of each member, Importance of Project site. Project contracts: Types and its contents. CPM & PERT Techniques.

UNIT III: Cost Behavior and Profit Planning: Marginal Costing, Distinction between Marginal Costing and Absorption Costing; Break-even Analysis, Cost-Volume-Profit Analysis, Standard Costing and Variance Analysis.

UNIT IV: Quality management and Budgeting strategies: Pareto Analysis, Target costing, Life Cycle Costing, Just-in-time approach, Material Requirement Planning, Enterprise Resource Planning, Total Quality Management and Theory of constraints. Activity-Based Cost Management, Bench Marking; Balanced Score Card and Value-Chain Analysis. **Budgetary Control;** Flexible Budgets; Performance budgets; Zero-based budgets. Measurement of Divisional profitability pricing & decisions including transfer pricing.

UNIT V: Quantitative techniques for cost management, Linear Programming, Transportation problems, Assignment problems, Simulation, Learning Curve Theory

Reference Books:

1. Cost Accounting A Managerial Emphasis, Prentice Hall of India, New Delhi
2. Charles T. Horngren and George Foster, Advanced Management Accounting
3. Robert S Kaplan Anthony A. Alkinson, Management & Cost Accounting
4. Ashish K. Bhattacharya, Principles & Practices of Cost Accounting A. H. Wheeler publisher
5. N.D. Vohra, Quantitative Techniques in Management, Tata McGraw Hill Book Co. Ltd.

Audit Course 1 &2

	English For Research Paper Writing	Course Code: V21PGENT54	L	T	P	C
			0	2	0	0

Course Outcomes: After completion of course, students would be able to

- CO1:** Present planning and preparation for breaking up long sentences by following word order and structuring paragraphs and sentences to avoid ambiguity and vagueness. **(K1)**
- CO2:** Clarify his/her findings by criticizing, hedging and paraphrasing to avoid plagiarism in writing the sections of the paper. **(K2)**
- CO3:** Construct the ability to review literature, methods, results, discussions and the final check. **(K3)**
- CO4:** Develop the key skills needed to write Title, Abstract, Introduction and Review of literature for a research paper. **(K3)**
- CO5:** Demonstrate the skills needed to write methods, results, the discussion and conclusions for Research Write-ups. **(K3)**
- CO6:** Employ useful phrases that ensure a paper for the first-time publication. **(K3)**

Syllabus:

Units	Contents	Hours
1	Planning and Preparation, Word Order, Breaking up Long sentences, Structuring Paragraphs and Sentences, Being Concise and Removing Redundancy, Avoiding Ambiguity and Vagueness.	4
2	Clarifying Who Did What, Highlighting Your Findings, Hedging and Criticising, Paraphrasing and Plagiarism, Sections of a Paper, Abstracts. Introduction	4
3	Review of the Literature, Methods, Results, Discussion, Conclusions, The Final Check	4
4	Key skills are needed when writing a Title, key skills are needed when writing an Abstract, key skills are needed when writing an Introduction, skills needed when writing a Review of the Literature.	4
5	Skills are needed when writing the Methods, skills needed when writing the Results, skills are needed when writing the Discussion, skills are needed when writing the Conclusions.	4
6	Useful phrases, how to ensure paper is as good as it could possibly be the first-time submission.	4

Suggested Studies:

1. Goldbort R (2006) Writing for Science, Yale University Press (available on Google Books)
2. Day R (2006) How to Write and Publish a Scientific Paper, Cambridge University Press
3. Highman N (1998), Handbook of Writing for the Mathematical Sciences, SIAM. Highman's book.
4. Adrian Wallwork, English for Writing Research Papers, Springer New York Dordrecht Heidelberg London, 2011

	Disaster Management	Course Code: V21STEAC1	L 0	T 2	P 0	C 0
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Course Outcomes: After completion of course, students would be able to

- CO1:** Describe to student to have a idea on different natural hazards and disaster management. **(K2)**
CO2: Develop the student to understand manmade disaster and their management. **(K3)**
CO3: Prepare the student in such a way inorder to understand building codes and vulnerability of disaster. **(K3)**
CO4: Illustrate to student about role of technology in disaster management. **(K2)**
CO5: Assess the importance of education and community preparedness in disaster management to student. **(K3)**

Syllabus:

UNIT I

Natural Hazards and Disaster Management: Introduction of DM Disaster Management cycle – Five priorities for action- Case study methods of the following: floods, droughts – Earthquakes – global warming, cyclones & Tsunamis – Post Tsunami hazards along the Indian coast – landslides.

UNIT II

Man Made Disaster And Their Management Along With Case Study Methods Of The Following: Fire hazards – transport hazard dynamics – solid waste management – post disaster – bio terrorism - rail and air craft's accidents-Management of these disasters

UNIT III

Risk And Vulnerability: – Building codes and land use planning – social vulnerability – environmental vulnerability -Financial management of disaster.

UNIT IV

Role Of Technology In Disaster Managements: Disaster management for infra structures, taxonomy of infra structure - mitigation programme for earth quakes –geospatial information in agriculture drought assessment-multimedia technology in disaster risk management and training- transformable indigenous knowledge in disaster reduction.

UNIT V

Education And Community Preparedness: Education in disaster risk reduction-Essentials of school disaster education-Community capacity and disaster resilience-Community based disaster recovery -Community based disaster management and social capital-Designing resilience- building.

Text Books:

1. Disaster Management – Global Challenges and Local Solutions' by Rajib shah & R R Krishnamurthy(2009),Universities press.
2. Disaster Science & Management' by Tushar Bhattacharya, Tata McGraw Hill Education Pvt. Ltd., New Delhi.
3. Disaster Management – Future Challenges and Opportunities' by Jagbir Singh (2007), I K International Publishing House Pvt. Ltd.

Reference Books:

1. 'Disaster Management' edited by H K Gupta (2003), Universities press.
2. Natural Hazards and Disaster Management, Vulnerability and Mitigation by RB Singh
3. Disaster Management by Harish K.Gupta

	Value Education	Course Code:	L	T	P	C
		V21PGENT55	0	2	0	0

Course Outcomes: After completion of course, students would be able to

CO1: Enumerate the societal values and Individual attitudes that lead to value based judgments. **(K1)**

CO2: Explain the need for value education that incorporates self-discipline, confidence, honesty and patriotism. **(K2)**

CO3: Develop the inner and external personality that transforms individual into a man of character. **(K3)**

CO4: Distinguish between character and competence, self-management and good health, mind your mind and self-control. **(K4)**

Syllabus:

Unit		Content	Hours
1	·	Values and self-development –Social values and individual attitudes.	4
	·	Work ethics, Indian vision of humanism.	
	·	Moral and non- moral valuation. Standards and principles.	
	·	Value judgment's	
2	·	Importance of cultivation of values.	6
	·	Sense of duty. Devotion, Self-reliance. Confidence, Concentration.	
	·	Truthfulness, Cleanliness.	
	·	Honesty, Humanity. Power of faith, National Unity.	
	·	Patriotism. Love for nature ,Discipline	
3	·	Personality and Behavior Development - Soul and Scientific attitude.	6
	·	Positive Thinking. Integrity and discipline.	
	·	Punctuality, Love and Kindness.	
	·	Avoid fault Thinking.	
	·	Free from anger, Dignity of labour.	
	·	Universal brotherhood and religious tolerance.	
	·	True friendship.	
	·	Happiness Vs suffering, love for truth.	
	·	Aware of self-destructive habits.	
	·	Association and Cooperation.	
4	·	Character and Competence –Holy books vs Blind faith.	6
	·	Self-management and Good health.	
	·	Science of reincarnation.	
	·	Equality, Nonviolence ,Humility, Role of Women.	
	·	All religions and same message.	
	·	Mind your Mind, Self-control.	
	·	Honesty, Studying effectively	

Suggested Reading:

1 Chakroborty, S.K. "Values and Ethics for organizations Theory and practice", Oxford University Press, New Delhi.

	Stress Management By Yoga	Course Code: V21PGENT53	L 0	T 2	P 0	C 0
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Course Outcomes: After completion of course, students would be able to

- CO1:** Define 8 parts of Yoga (Ashtanga). (K1)
- CO2:** Discuss Yam and Niyam along with Dos and Don'ts in life. Interpret Ahimsa, satya, astheya, brahmacharya and aparigraha along with other concepts. (K2)
- CO3:** Practice Asan and Pranayam. Examine various yoga poses and their benefits for mind and body. (K3)

Syllabus:

Unit	Content	Hours
1	Definitions of Eight parts of yoga. (Ashtanga)	8
2	Yam and Niyam. Do's and Don'ts in life. i) Ahimsa, satya, astheya, brahmacharya and aparigraha ii) Shaucha, santosh, tapa, swadhyay, ishwarpranidhan	8
3	Asan and Pranayam 1. Various yog poses and their benefits for mind & body 2. Regularization of breathing techniques and its effects-Types of pranayam	8

Suggested Readings:

1. "Yogic Asanas for Group Training-Part-I": Janardan Swami Yogabhyasi Mandal, Nagpur
2. "Rajayoga or conquering the Internal Nature" by Swami Vivekananda, Advaita Ashrama (Publication Department), Kolkata

	Personality Development through Life Enlightenment Skills	Course Code: V21PGENT52	L 0	T 2	P 0	C 0
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Course Outcomes: After completion of course, students would be able to

CO1: Relate Neetishatakam in developing versatile personality of students. **(K1)**

CO2: Employ Bhagavad Gita to lead the nation and mankind to peace and prosperity. **(K3)**

CO3: Connect students to Bhagavad Gita in order to develop personality and achieve highest goals in life. **(K4)**

Unit	Content	Hours
1	Neetisatakam-Holistic development of personality	8
	· Verses- 19,20,21,22 (wisdom)	
	· Verses- 29,31,32 (pride & heroism)	
	· Verses- 26,28,63,65 (virtue)	
	· Verses- 52,53,59(dont"s)	
	· Verses- 71,73,75,78(do"s)	
2	· Approach to day to day work and duties.	8
	· Shrimad Bhagavad Gita: Chapter 2-Verses 41, 47,48,	
	· Chapter 3-Verses 13, 21, 27, 35, Chapter 6-Verses 5,13,17, 23, 35,	
	· Chapter 18-Verses 45, 46, 48.	
3	· Statements of basic knowledge.	8
	· Shrimad Bhagwad Geeta: Chapter2-Verses 56, 62, 68	
	· Chapter 12 -Verses 13, 14, 15, 16,17, 18	
	· Personality of Role model. Shrimad Bhagwad Geeta: Chapter2-	
	Verses 17, Chapter 3-Verses 36,37,42,	
	· Chapter 4-Verses 18, 38,39	
	· Chapter18 – Verses 37,38,63	

Suggested reading

1. "SrimadBhagavadGita"bySwamiSwarupananda Advaita Ashram (Publication Department), Kolkata
2. Bhartrihari"s Three Satakam (Niti-sringar-vairagya) by P.Gopinath, Rashtriya Sanskrit Sansthanam, New Delhi.

	Pedagogy Studies	Course Code: V21PGENT51	L 0	T 2	P 0	C 0
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Course Outcomes: After completion of course, students would be able to

- CO1:** Identify various theories of learning and recognize Research questions with an overview of methodology and searching. (K2)
- CO2:** Review Pedagogical practices used by teacher in both formal and informal classroom and design Curriculum. (K2)
- CO3:** Examine how teacher education and the school curriculum support effective pedagogy along with various pedagogical approaches and theories. (K3)
- CO4:** Show peer support for professional development and support from head teacher to develop curriculum and assessment. Find out the barriers involved in learning. (K3)
- CO5:** Find out the gaps and give directions for research design as per context. (K3)

Syllabus:

Units	Content	Hours
1	Introduction and Methodology: • Aims and rationale, Policy background, Conceptual framework and terminology • Theories of learning, Curriculum, Teacher education. • Conceptual framework, Research questions. • Overview of methodology and Searching.	4
2	• Thematic overview: Pedagogical practices are being used by teachers in formal and informal classrooms in developing countries. • Curriculum, Teacher education.	2
3	• Evidence on the effectiveness of pedagogical practices • Methodology for the in depth stage: quality assessment of included studies • How can teacher education (curriculum and practicum) and the school curriculum and guidance materials best support effective pedagogy? • Theory of change. • Strength and nature of the body of evidence for effective pedagogical practices. • Pedagogic theory and pedagogical approaches. • Teachers' attitudes and beliefs and Pedagogic strategies.	4
4	• Professional development: alignment with classroom practices and follow-up support • Peer support • Support from the head teacher and the community. • Curriculum and assessment • Barriers to learning: limited resources and large class sizes	4
5	Research gaps and future directions • Research design • Contexts • Pedagogy • Teacher education • Curriculum and assessment • Dissemination and research impact	2

Suggested reading

1. Ackers J, Hardman F (2001) Classroom interaction in Kenyan primary schools, *Compare*, 31 (2): 245-261.
2. Agrawal M (2004) Curricular reform in schools: The importance of evaluation, *Journal of Curriculum Studies*, 36 (3): 361-379.
3. Akyeampong K (2003) Teacher training in Ghana - does it count? Multi-site teacher education research project (MUSTER) country report 1. London: DFID.
4. Akyeampong K, Lussier K, Pryor J, Westbrook J (2013) Improving teaching and learning of basic maths and reading in Africa: Does teacher preparation count? *International Journal Educational Development*, 33 (3): 272-282.
5. Alexander RJ (2001) *Culture and pedagogy: International comparisons in primary education*. Oxford and Boston: Blackwell.
6. ChavanM(2003)ReadIndia:Amassscale,rapid,,,learningtoread“campaign.
7. www.pratham.org/images/resource%20working%20paper%202.pdf.

1. What pedagogical practices are being used by teachers in formal and informal classrooms in developing countries?
2. What is the evidence on the effectiveness of these pedagogical practices, in what conditions, and with what population of learners?
3. How can teacher education (curriculum and practicum) and the school curriculum and guidance materials best support effective pedagogy?